Computer Networks 257 (2025) 110984

Contents lists available at ScienceDirect

Computer Networks

journal homepage: www.elsevier.com/locate/comnet

Survey paper

A survey on encrypted network traffic: A comprehensive survey of identification/classification techniques, challenges, and future directions

Adit Sharma *, Arash Habibi Lashkari

Behaviour-Centric Cybersecurity Center (BCCC), School of Information Technology, York University, Toronto, Ontario, Canada

ARTICLE INFO

Keywords:
Network traffic analysis
Encrypted traffic analysis
Encrypted traffic datasets
Network traffic analyzers
Encrypted traffic detection
Network traffic classification
Mobile traffic classification
ETC

ABSTRACT

Encrypted traffic detection and classification is a critical domain in network security, increasingly essential in an era of pervasive encryption. This survey paper delves into integrating advanced Machine Learning (ML) and Deep Learning (DL) techniques to address the challenges of robust encryption methods and dynamic network behaviors. Despite notable advancements, there remains a substantial gap in the operational application of these technologies, often constrained by scalability, efficiency, and adaptability to varied encryption standards. We critically review existing methodologies from 7 surveys and 82 related technical papers, highlight the shortcomings, and propose future research directions. Our analysis underscores the need to develop innovative, resource-efficient models that seamlessly adapt to new threats and encryption techniques without compromising performance. Additionally, we advocate for creating comprehensive datasets that merge encrypted and non-encrypted traffic to enhance model training and testing. This survey maps out the trajectory of recent developments and charts a course for future research that could significantly enhance encrypted traffic management and security capabilities.

1. Introduction

Encryption has become a cornerstone of data privacy in the digital age, with its adoption driven by escalating security concerns and stringent regulatory frameworks [1]. The rapid escalation of encrypted internet traffic has profoundly impacted network security dynamics, with recent data indicating that as of 2023, approximately 95% of web traffic is encrypted using HTTPS protocols [2]. This significant increase from previous years underscores a global shift towards prioritizing data privacy and security. However, this surge in encryption has introduced new challenges for network management, security monitoring, and quality of service, prompting the development of sophisticated Encrypted Traffic Analysis (ETA) techniques [3].

Cybercriminals increasingly leverage encrypted channels to conceal malicious activities, complicating traditional security measures. A 2023 report by Zscaler ThreatLabz revealed that 85.9% of cyberattacks now utilize encrypted channels, marking a 20% increase from the previous year [4]. This trend poses a challenge, as encrypted traffic can obscure threats from standard detection tools. Additionally, the adoption of TLS 1.3 has accelerated, with a notable rise in its implementation over the past two years. While TLS 1.3 enhances security and performance, it also limits visibility into network traffic, posing challenges for monitoring and threat detection [5].

In response to these developments, organizations increasingly invest in advanced security solutions that inspect encrypted traffic without compromising privacy. Techniques such as machine learning and artificial intelligence are employed to analyze traffic patterns and detect anomalies indicative of malicious behavior [6]. In summary, the exponential growth of encrypted traffic over the past few years has necessitated reevaluating network security strategies. Balancing the benefits of encryption with the imperative of effective threat detection remains a critical focus for researchers and practitioners.

This survey paper delves into the advancements in Machine Learning (ML) and Deep Learning (DL) models that are at the forefront of detecting and analyzing encrypted traffic [7]. These models offer promising solutions by enabling effective network monitoring and anomaly detection without decrypting the data, thus preserving user privacy [8]. We explore various ML approaches, including supervised and unsupervised techniques [9], and DL strategies such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which have shown considerable success in pattern recognition within encrypted flows [10].

In encrypted traffic detection and classification, network security can be broadly divided into two critical aspects: security and privacy. While security focuses on protecting the network infrastructure from

* Corresponding author.
E-mail address: adit27@yorku.ca (A. Sharma).

https://doi.org/10.1016/j.comnet.2024.110984
Received 3 September 2024; Received in revised form 13 November 2024; Accepted 5 December 2024
Available online 15 December 2024

unauthorized access, attacks, and vulnerabilities [11], privacy emphasizes safeguarding the confidentiality of the data being transmitted within that network [12]. Both aspects are vital for ensuring the overall integrity of network communications, especially in today's digital landscape where large volumes of sensitive information are constantly in transit [13].

Encryption techniques have recently been enhanced by integrating blockchain technology, which offers a decentralized and transparent platform that significantly strengthens the security of IoT devices—integral components in managing encrypted traffic [14]. Blockchain's ability to maintain immutable records and provide enhanced privacy through advanced encryption techniques has become crucial in ensuring security and privacy within network communications [15]. Additionally, the convergence of blockchain with IoT fortifies the security landscape while introducing innovative privacy-preserving strategies that protect data across distributed networks [16]. These advancements are instrumental in developing encrypted traffic detection and analysis, offering robust solutions against potential cyber threats [17,18].

Focusing on privacy within industrial IoT applications, secure inference solutions such as Secure Multiparty Computation (SMC) have emerged as effective tools to mitigate privacy risks. These solutions allow multiple parties to securely conduct AI reasoning without exposing sensitive data, making them particularly valuable in environments like smart manufacturing [11]. Privacy-enhancing techniques, such as Geo-Indistinguishability—demonstrated through Group-Based Noise Addition (CANOE)—protect individual privacy in spatial crowdsourcing, ensuring efficient task allocation without compromising the utility of spatial data [12]. Moreover, tensor-based Recurrent Neural Networks (RNNs) integrated with differential privacy address privacy concerns in IoT environments by securely processing heterogeneous sequential data, maintaining confidentiality across various IoT platforms [13]. These methodologies contribute to advancing privacy-preserving technologies and play a pivotal role in enhancing encrypted traffic analysis in complex network environments.

Further enhancing the dialogue on encrypted network traffic, recent studies delve into the dynamics of Tor networks, emphasizing the dual challenge of ensuring user anonymity and system integrity. Notably, Shahbar and Zincir-Heywood [19] investigates traffic flow analysis within Tor's pluggable transports, revealing vulnerabilities that could potentially be exploited to compromise user privacy. Concurrently, Shahbar and Zincir-Heywood [20] benchmarks methods for classifying Tor traffic at both flow and circuit levels, illustrating the complexities of discerning user activities without decrypting traffic, thereby underscoring the intricate balance between operational transparency and user confidentiality. Additionally, Montieri et al. [21] and Montieri et al. [22] explore the efficacy of traffic classification techniques across various anonymity tools, including Tor, I2P, and JonDonym, highlighting the significant challenges in identifying the type of traffic and the applications it encapsulates due to the robust encryption and routing mechanisms employed. These insights illuminate the ongoing tensions between enhancing security measures and preserving the stringent privacy requirements demanded in modern networked environments.

This survey paper seeks to paint a broad picture of ETA's current landscape and emerging trends by providing a comprehensive overview of the methods, challenges, and datasets integral to this domain.

The structure of the survey paper is as follows. The second section, the Search Methodology, outlines the systematic approach used to select and analyze the technical articles and datasets that form the foundation for this survey.

The third section, Previous Survey Papers & Motivation, reviews the existing literature, contextualizing this survey within the broader research landscape. It also discusses the compelling reasons for undertaking this survey, including the rapid evolution of encryption techniques and their implications on network security.

The fourth and fifth sections, Encryption Traffic Techniques/ Services and Related Work - Technical Articles, explore the encryption landscape. The fourth section details the prevalent encryption protocols and services, laying the groundwork for understanding their detection and analysis. Subsequently, the fifth section delves into the core of recent research, categorizing efforts into machine learning (ML), deep learning (DL) models, and hybrid approaches utilized in encrypted traffic detection.

The sixth and seventh sections, titled Encrypted Traffic Datasets and Information Extractors (IE) or Traffic Analyzers, offer an in-depth exploration of the resources and tools pivotal to encrypted traffic analysis. The sixth section critically examines the available datasets for training and testing encrypted traffic detection systems. The seventh section then analyzes the tools and techniques developed to extract actionable insights from encrypted data, ensuring privacy concerns are respected.

The eighth and ninth sections, titled Challenges & Future Work and Conclusion, respectively, conclude the survey by addressing the landscape of encrypted traffic analysis. The eighth section highlights the unresolved issues and potential research directions that could address the limitations of current methods. Finally, the ninth section synthesizes the findings from each section, emphasizing the integration of machine learning techniques with encrypted traffic analysis and outlining the implications for both research and practical applications.

The Fig. 1 succinctly outlines the logical progression and structure of our survey paper. It starts with an introductory overview, followed by a detailed search methodology, leading to discussions on previous surveys and the motivation for a new analysis. This flowchart then delineates the path through various sections of the paper, including encryption techniques, relevant technical articles, and a synthesis of machine learning, deep learning, and hybrid approaches. It continues through the evaluation of encrypted traffic datasets and analyzers, culminating in a discussion on current challenges and future directions before concluding. This structural diagram enhances the reader's comprehension of the paper's framework, providing a clear visual guide through the intricate landscape of encrypted traffic classification research.

The acronyms and abbreviations used throughout this paper are comprehensively detailed in Table 12, located in Appendix. This table provides a clear reference for understanding all technical terms utilized in the study.

This survey paper has endeavoured to provide a comprehensive overview and a structured analysis of encrypted traffic detection and classification. The significant contributions of this work can be summarized as follows:

- **Taxonomy of Encrypted Traffic Detection and Classification Models:** We have developed a detailed taxonomy that categorizes the existing models based on their methodological approaches and operational capabilities, offering a clear framework for understanding the landscape of current technologies.
- **Survey of Techniques for Encrypted Traffic Detection and Classification:** This paper reviews various techniques and methodologies employed in encrypted traffic detection, providing insights into their effectiveness and application scenarios.
- **Comprehensive Analysis of Datasets:** We compile and present a table summarizing all available datasets pertinent to encrypted traffic studies, listing their features and characteristics. This compilation is a valuable resource for researchers to evaluate and select appropriate study datasets.
- **Overview of Technical Analyzers/Information Extractors:** A comprehensive table is introduced, summarizing all existing analyzers and information extractors, highlighting their features and utility. This overview facilitates a better understanding of the tools for integrating into encrypted traffic detection systems.

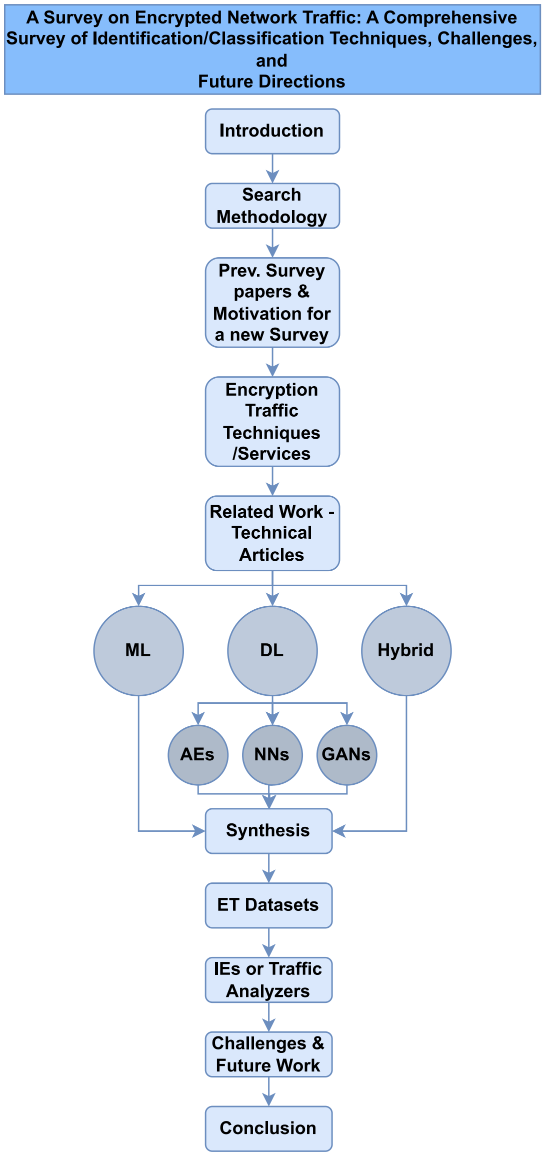

Fig. 1. Comprehensive flowchart of methodological approach in encrypted traffic detection and classification.

- **Proposed Future Research Directions:** Based on the gaps identified through our survey, we outline potential future research directions that could address existing challenges and advance the encrypted traffic detection and classification field.

As encrypted traffic becomes increasingly prevalent, the necessity for robust ETA methods that respect privacy while ensuring network security is more critical than ever [23]. This survey paper serves as a primer on the state-of-the-art ML and DL techniques shaping the future of encrypted network traffic analysis. Through a detailed review of the technological advancements and their applications, we aim to foster a deeper understanding of the field and spur further innovation in the ongoing battle between data privacy and network security.

2. Search methodology

This section presents an investigation methodology for conducting a comprehensive encrypted traffic detection and classification search. This field includes domains such as ML, DL, and Hybrid approaches.

Table 1
List of digital libraries utilized in ETC research.

Site	URL
Springer	link.springer.com
Elsevier	sciencedirect.com
Wiley	onlinelibrary.wiley.com
IEEE explorer	ieeexplore.ieee.org
ACM Digital Library	dl.acm.org
Cornell University Archive	arxiv.org
SSRN	papers.ssrn.com
MDPI	mdpi.com
SSRN	papers.ssrn.com
Spied Digital	spieddigitallibrary.org

Fig. 2 presents the five steps of the proposed search methodology, beginning with selecting keywords and search items for various scientific publishers.

In the first step, the following search queries and keywords were employed across various scientific publishers (Listed in Table 1) to identify existing survey papers :

- Encrypted Traffic Survey
- Encrypted Traffic Review
- Encrypted Traffic Classification Overview
- Encrypted Traffic Classification Survey
- Encrypted Traffic Classification Review
- Encrypted Traffic Classification using ML Approaches Survey
- Encrypted Traffic Classification using ML Approaches Review
- Encrypted Traffic Classification using DL Approaches Review
- Encrypted Traffic Classification using DL Approaches Survey

For the technical articles, the following keywords were used:

- Encrypted, Network, Traffic, Classification, Detection, Characterization, ML, DL, Feature Extraction, Supervised Learning, Unsupervised Learning, Neural Networks, Artificial Intelligence, Internet Security, Cybersecurity etc.

In the second step, we collect all relevant literature comprehensively, spanning review articles and technical papers published within the last five years. This aggregation is sourced from databases and journals, as delineated in Table 1. This step ensures the inclusion of recent advancements and contemporary research findings, thus laying a foundation for a robust literature analysis.

The selected articles are checked for the third step to eliminate duplicates and repeated studies. This process utilizes manual review to ensure that each included article presents unique insights and contributes substantively to the breadth of knowledge being surveyed. This refinement paves the way for a focused and concise synthesis of literature.

Step four entails creating a unique list of articles by excluding those irrelevant to the core topics of interest. This step includes a feedback loop to step three, allowing for refinement and reevaluation of article relevance, ensuring that only the most pertinent and significant articles are considered for in-depth review.

The final step of our methodology involves generating a definitive list of technical and survey articles. This culmination involves a thorough review, which includes a validation check to confirm the relevance and credibility of the sources. The resulting compilation is poised for in-depth analysis and synthesis, providing an extensive summary of the present status of research as it pertains to our survey topic.

3. Previous survey papers & Motivation for a new survey paper

Cybersecurity has seen a marked increase in the significance of detecting and classifying encrypted traffic, driven by the dual needs

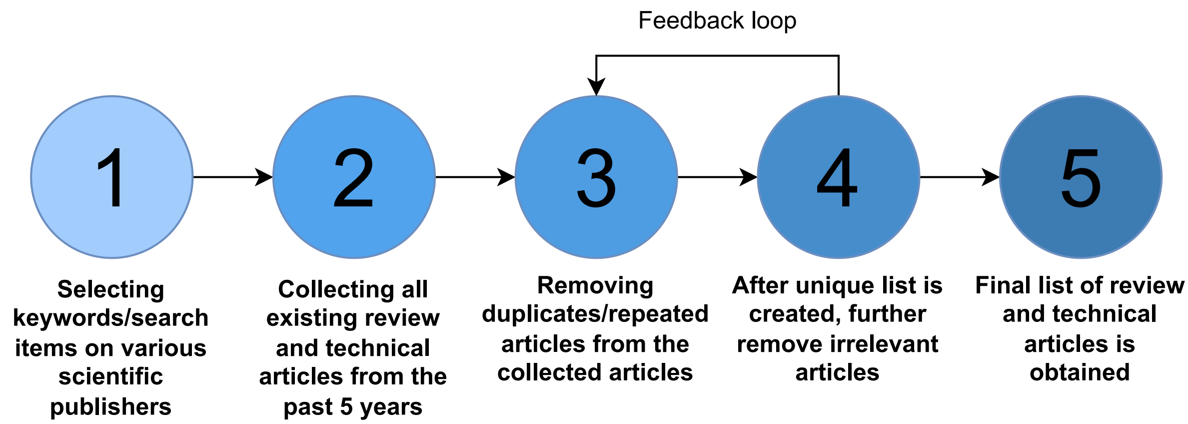

Fig. 2. Structured overview of methodological steps in encrypted traffic detection and Classification Survey.

of privacy and security [7]. Encrypted traffic, which includes protocols like TLS, SSL, and QUIC, now forms a substantial segment of global internet communications [24]. While encryption protects data from unauthorized access, it also complicates network management and anomaly detection [3]. As detailed in the previous section of this survey paper, a meticulously crafted search methodology was employed to address these challenges. This approach enabled the systematic review and selection of survey papers that span the full spectrum of encrypted traffic analysis techniques from the past decade in a chronological manner.

The survey papers reviewed here build upon one another, tracing the evolution from initial methods such as Deep Packet Inspection (DPI) to sophisticated, privacy-preserving ML algorithms [8]. They reflect a broader shift towards techniques that maintain user privacy while providing security professionals with the necessary tools to monitor and respond to threats effectively [10]. This section explores these key contributions to the field and presents a thorough overview of existing survey papers on the classification and detection of encrypted traffic.

The study by Velan, P. et al. provides an in-depth review of methods for classifying and analyzing encrypted network traffic, examining prevalent encryption protocols and their impact on network security. The paper delineates a range of classification techniques, from payload-based to feature-based methods, and organizes them into a structured taxonomy. It presents an overview of how the initiation phase of encrypted connections can offer valuable information for enforcing security policies, highlighting how certain unencrypted elements, such as protocol versions and cipher suites, can be exploited for network anomaly detection and client identification. The survey emphasizes the intricacy of classifying encrypted traffic due to varied protocol behaviors and the importance of selecting appropriate classification methods that do not compromise user privacy. This work is especially relevant given the increasing use of encryption, which complicates traditional traffic analysis. The authors underscore the evolution of classification strategies to effectively address the challenges posed by encrypted communications, emphasizing the need for advanced techniques that can adapt to the enhanced security measures of modern network traffic. The survey is a vital reference for the ongoing adaptation and innovation in encrypted traffic analysis, signaling the significant impact of encryption on network neutrality and end-user privacy [25].

Aminuddin, M. A. I. M. et al. propose a survey on ML methods for monitoring encrypted Tor traffic. The paper reviews various ML strategies, including supervised, semi-supervised, and unsupervised learning, each contributing distinct advantages to analyzing encrypted network traffic. The authors discuss the efficacy of conventional algorithms, including Decision Trees, Support Vector Machines, and Naive Bayes, successfully applied to classify patterns in encrypted Tor traffic. Additionally, the survey highlights the use of advanced algorithms for identifying specific traffic types within Tor, such as pluggable transports or distinguishing between different network applications. Building on these insights, the paper also addresses the broader challenges and considerations in classifying encrypted traffic using ML approaches, noting the importance of dataset and feature selection, the challenges posed by the encryption's complexity, and the need for real-time classification capabilities. It emphasizes that no single algorithm can perform optimally under all conditions, highlighting the relevance of selecting appropriate ML models based on accuracy, training time, computational resources, and feature selection to enhance privacy protection within the Tor network. The comprehensive nature of this survey underscores the significant progress in applying ML to bolster the security and analysis of encrypted network traffic within anonymity networks like Tor [26].

Rezaei et al. and Iliyasu et al. provide comprehensive analyses of DL techniques in ETC. Rezaei et al. focus on the utility of CNNs for their hierarchical processing capabilities and delve into the application of Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks(RNNs) for handling sequential data integral to network traffic flows. They highlight the growth of hybrid models that blend these neural architectures to improve the accuracy of classification in the face of encryption protocols like QUIC and TLS 1.3, as well as complex challenges such as multi-label classification in multiplexed streams [27].

Iliyasu et al. echo this narrative by underscoring the importance of LSTM and CNN networks, especially in capturing long-term dependencies in encrypted traffic data. Their review extends into the realm of hybrid DL models, which they present as advanced, versatile tools capable of addressing the limitations presented by increasingly sophisticated encryption methods. Both surveys note significant hurdles, such as gathering and labeling comprehensive datasets and managing multiplexed traffic streams that combine multiple traffic classes within single flows. Collectively, these works illustrate a landscape where DL enhances traffic analysis capabilities and faces a continuous challenge from the evolving scope of network encryption [28].

Li, Y. et al. propose a comprehensive survey that underscores the critical role of DL and ML in detecting encrypted malicious traffic. The authors meticulously examine the application of CNNs for robust feature extraction from encrypted data, which is foundational in identifying potential security threats. The paper further explores the significance of sequence data processing capabilities of RNNs and LSTMs, which are instrumental for maintaining contextual awareness in traffic flow analysis. The authors also shed light on hybrid DL architectures, which amalgamate the strengths of individual models to yield a more sophisticated and accurate detection mechanism. The survey concludes that these advanced methodologies signify a paradigm shift in network security strategies, moving away from traditional heuristic-based systems towards more dynamic, data-driven models that can keep pace with the complexities introduced by modern encryption standards. Moreover, the paper discusses ongoing issues and challenges, such as the scarcity of public datasets, the necessity for more robust feature extraction techniques, and the difficulty of implementing these technologies in real networks with considerable variability. It emphasizes that while the identification of encrypted malicious traffic has progressed, transitioning from passive to active defense mechanisms remains a pivotal area for future research, suggesting an imminent need to improve the applicability and real-world efficacy of current detection technologies [29].

In [30], Papadogiannaki, E., & Ioannidis, S. conduct a thorough investigation into the domain of encrypted network traffic analysis in their paper 'A Survey on Encrypted Network Traffic Analysis: Applications, Techniques, and Countermeasures,' published in ACM Computing Surveys (CSUR). This survey navigates through the increasing necessity for sophisticated methods to analyze and classify encrypted traffic, driven by the widespread adoption of encryption protocols aimed at enhancing security and protecting user privacy. The authors meticulously categorize the spectrum of research efforts into applications for network analytics, security enhancements, and privacy protections. They highlight the significant challenges around dataset construction, traffic representation, and analysis model building, as outlined in their review. The survey emphasizes the instrumental role of ML and DL in deciphering encrypted traffic nuances without undermining the core advantages of encryption, spotlighting the need for innovative solutions that reduce false positives and improve real-time processing performance in encrypted networks. By setting a foundation for advancing this critical area of network analysis and security, Papadogiannaki and Ioannidis call for directions of future research that address these ongoing challenges, enriching the field with new methodologies and tools.

In [31], Shen, M. et al. present a comprehensive survey titled 'Machine Learning-Powered Encrypted Network Traffic Analysis' in IEEE Communications Surveys & Tutorials. This extensive study encapsulates the evolution of encrypted traffic analysis from traditional methodologies, which struggle with the advent of encryption, to modern ML techniques that proficiently unravel encrypted data's complexities. The survey highlights the significant challenges in dataset construction, traffic representation, analysis model building, and developing countermeasures against encryption while also offering a detailed taxonomy of analysis objectives such as anomaly detection and privacy leakage. It reviews the diverse machine-learning tools at the forefront of this field. It emphasizes the necessity for high-quality datasets, robust traffic representations, and adaptive analysis models to enhance accuracy and efficiency. By underlining the importance of addressing these challenges, the survey guides future research towards developing more robust and flexible security frameworks, thereby adapting measures for network security to the evolving landscape of widespread encryption.

3.1. Motivation

The comprehensive review, systematically summarized in Table 2, uncovers notable limitations in the methodologies and the scope of existing surveys on encrypted traffic classification. Our analysis reveals that many of these surveys tend to focus narrowly on specific segments of the field, often emphasizing machine learning (ML) and deep learning (DL) applications [25,26]. This table highlights the areas where these studies lack breadth or fail to capture emerging technologies and approaches beyond the conventional ML and DL paradigms. It also sheds light on the underrepresentation of hybrid and advanced analytical models that combine multiple techniques for enhanced accuracy and efficiency. However, they often overlook the integration of these technologies into real-world scenarios, resulting in gaps such as inadequate protocol-specific analyses, insufficient adaptability of feature sets across varied encryption standards, and a general disregard for operational challenges like scalability and efficiency in deployment [28,31].

Moreover, the fragmented approach observed in existing studies concerning the impact of advanced encryption on the classification of traffic methods hampers the accurate classification and detection of encrypted traffic [29]. There is a clear need for a survey consolidating existing knowledge and addressing areas poorly covered or omitted in prior surveys.

By pinpointing these gaps, Table 2 serves as a critical resource for guiding future research towards more comprehensive and inclusive studies that address the evolving complexities of encrypted traffic detection.

In light of these observations, this survey seeks to bridge identified gaps by integrating upcoming ML/DL models and privacy-enhancing technologies. This approach is vital for developing solutions that meet the increasingly complex demands of modern cybersecurity. This survey aims to analyze encrypted network traffic thoroughly, employing advanced ML and DL techniques alongside an in-depth examination of the evolving landscape of encryption protocols. Such an analysis will equip researchers and practitioners with a robust toolkit for navigating the complexities of encrypted traffic analysis, addressing theoretical advancements and practical deployment challenges.

The primary objective of this survey is to address the following questions regarding the different detection and classification approaches proposed for encrypted traffic:

Fundamental Research Questions:

- Question 1: "Has there been a comprehensive taxonomy of the various data collection approaches utilized in the domain of encrypted traffic analysis? - Covered in Section 5 - Related Work - Technical Articles "
- Question 2: "Are the encryption protocols used for secure data transmission adequately classified and analyzed within existing literature? - Covered in Section 4 - Encryption Traffic Techniques/Services"

Technical Questions:

- Question 3: "What information extraction methods are employed throughout the analysis, and are these thoroughly surveyed? - Covered in Section 7 - Information Extractors (IE) or Traffic Analyzers"
- Question 4: "How extensively have traffic classification techniques been explored? - Covered in Section 5 - Related Work - Technical Articles"
- Question 5: "What ML models are applied and reviewed? - Covered in Section 5 - Related Work - Technical Articles"
- Question 6: "To what extent have DL techniques been utilized, and is there a detailed survey of their effectiveness? - Covered in Section 5 - Related Work - Technical Articles"
- Question 7: "Are the features of encrypted traffic and their classification methods systematically cataloged in the surveyed papers? - Covered in Section 5 - Related Work - Technical Articles"
- Question 8: "Is there an assessment of the performance metrics & datasets utilized in the surveyed studies? - Covered in Sections 5 & 6 - Related Work - Technical Articles & Encrypted Traffic Datasets respectively"
- Question 9: "How have model selection techniques been incorporated, and are they evaluated comprehensively? - Covered in Section 5 - Related Work - Technical Articles"

Research Gaps & Future Work Questions:

- Question 10: "How is encrypted traffic detection technology covered in existing research, and what gaps remain? - Covered in Sections 5 & 8 - Related Work - Technical Articles & Challenges & Future Work respectively"

Table 2
Comparative survey of techniques employed for encrypted traffic classification and analysis where covered ✓✓; partially covered ✓; not covered ✗.

Title	Year	Area covered	Taxonomy	Data collection techniques	Encryption protocols	Information/Feature extraction	Traffic classification techniques	ML techniques	DL techniques	Performance evaluation metrics	Model selection techniques
A survey of methods for ETC and analysis [25]	2015	Encrypted traffic identification	✓✓	✗	✓✓	✓✓	✓✓	✓✓	✗	✗	✗
A Survey on Tor Encrypted Traffic Monitoring [26]	2018	Tor/non-tor encrypted traffic classification	✓✓	✗	✓✓	✗	✓✓	✓✓	✗	✗	✗
Deep Learning for Encrypted Traffic Classification: An Overview [27]	2019	DL for encrypted traffic classification	✓✓	✓✓	✗	✗	✗	✓	✓✓	✗	✓✓
A Review of Deep Learning Techniques for Encrypted Traffic Classification [28]	2020	Deep Learning Techniques for Encrypted Traffic Classification	✓✓	✗	✗	✗	✗	✗	✓✓	✗	✗
A Survey of Encrypted Malicious Traffic Detection [29]	2021	Encrypted traffic identification	✗	✓✓	✓✓	✓✓	✗	✓✓	✓	✓✓	✓
A Survey on Encrypted Network Traffic Analysis Applications, Techniques, and Countermeasures [30]	2021	Encrypted traffic in network analysis	✓✓	✗	✗	✗	✓✓	✓✓	✓✓	✓✓	✓✓
Machine Learning-Powered Encrypted Network Traffic Analysis [31]	2022	Encrypted traffic in network analysis	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓	✓✓	✓✓
This paper	**2024**	**ETC Techniques, Challenges, and Future Directions**	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓	✓✓

The subsequent sections of this survey paper focus on covering the questions mentioned above in detail and elaborating on these methodologies, providing a more informed and effective approach to encrypted traffic detection and classification techniques. This comprehensive analysis is essential in an era characterized by rapidly evolving encryption standards and protocols, underscoring the evolving nature of the cybersecurity field and its critical role in maintaining the integrity of online communications.

4. Encryption traffic techniques/services

Encryption techniques and services have evolved significantly over the decades, responding to the growing demand for secure communication across various domains, including military, commercial, and personal interactions [32]. At its core, encryption converts plaintext into ciphertext using an algorithm and a key, rendering the data unreadable to unauthorized parties [33]. Historically, encryption began with simple ciphers, such as the Caesar cipher, which employed a basic substitution method [32]. However, as computational power increased, encryption methods became more sophisticated, creating complex algorithms that could withstand advanced cryptanalysis techniques [34].

The trajectory of cryptography, from these simple methods to advanced systems, mirrors the continuous development of robust encryption standards like the Data Encryption Standard (DES) and the Advanced Encryption Standard (AES), designed to resist computational attacks. Modern techniques such as differential privacy and homomorphic encryption represent the cutting edge of this evolution. Differential privacy, for example, adds controlled random noise to data, providing strong privacy guarantees without significantly compromising data utility, which is particularly critical in areas like cloud computing and big data analytics [35]. Meanwhile, homomorphic encryption allows computations on encrypted data without decryption, ensuring secure data processing and maintaining privacy in scenarios such as cloud environments [36].

These advancements reflect cryptography's progression towards securing complex and large-scale digital communications, highlighting historical techniques and modern innovations that address today's security challenges.

This section meticulously examines and highlights the characteristics of various encryption techniques and services prevalent in the field. Tables 3, 4, 5, 6, and 7 collectively offer a comprehensive comparative analysis of various encryption services used in traffic management, including Tor, VPNs, and other emerging technologies. These tables detail the integration of these services with different analytical models, highlighting their effectiveness in enhancing privacy and security. Specifically, they examine the diverse approaches and algorithms used across different studies, providing insights into the strengths and weaknesses of each encryption technique. This collective analysis aids in understanding the broader implications of encryption technologies on the accuracy and reliability of traffic classification systems, thereby illuminating the specific challenges and benefits associated with each service.

Fig. 3 depicts the count of encryption protocols/services employed in the various technical articles reviewed. It shows the widespread use of protocols/services cited in various studies, highlighting the predominant use of VPNs, noted in 30 instances, underscoring their widespread acceptance for securing online communications. Tor usage, observed in 9 cases, and use of VPN and Tor combined in 6 instances reflect specialized approaches for enhancing user anonymity. The 'Others' category, comprising 27 instances, includes diverse technologies like I2P, Zeronet, and cryptographic protocols such as HTTPS and SSL, illustrating various encryption strategies catering to different security needs. This distribution invites further investigation into why certain protocols are preferred in specific scenarios and how combinations of these technologies might offer enhanced security. Such insights are necessary for developing advanced, user-friendly encryption solutions.

By examining these detailed comparisons, future researchers can discern patterns and effectiveness of different encryption models, assess the synergy between encryption methods and ML algorithms, and identify trends that may influence the direction of future innovations. Such insights are invaluable for scholars aiming to enhance existing encryption frameworks or to pioneer novel approaches that align with the evolving landscape of digital security technologies.

- **VPN (Virtual Private Network) (1996, Gurdeep Singh-Pall)** - VPNs create a private network from a public internet connection, encapsulating and encrypting data packets to provide secure and anonymous access to the internet. This technology is fundamental in bypassing geo-restrictions and safeguarding sensitive data from unauthorized access, especially beneficial in environments that require confidentiality like corporate networks [37].
- **TOR (The Onion Router) (2002, Paul Syverson, Michael G. Reed, David Goldschlag - U.S. Naval Research Laboratory)** — TOR is a network of servers that enables anonymous communication by directing internet traffic through a free, worldwide,

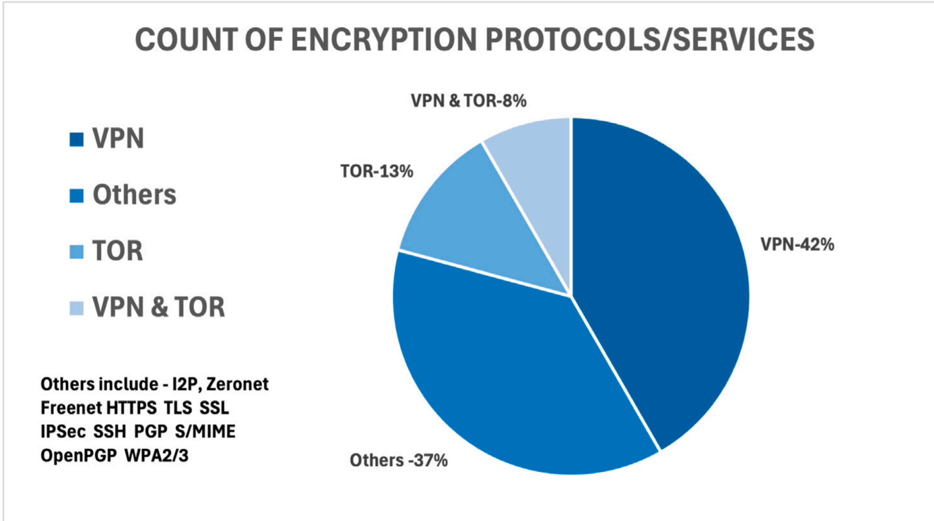

Fig. 3. Quantitative overview of encryption protocols and services in encrypted traffic studies.

volunteer overlay network consisting of more than seven thousand relays. The design ensures that the user's location and usage from anyone conducting network surveillance or traffic analysis is hidden, making it a crucial tool for preserving privacy and freedom online [38].

- **I2P (Invisible Internet Project) (2003, I2P Team and Community)** — I2P specializes in allowing anonymous communication over the internet. It uses a peer-to-peer-like routing structure to create a private network layer that enables secure and anonymous communication. Unique to I2P is its use of garlic routing, which encapsulates multiple messages together to enhance security and privacy beyond traditional onion routing [39].
- **Zeronet (2015, Tamas Kocsis)** — Zeronet uses Bitcoin cryptography and BitTorrent technology to build a decentralized web. It enables open, free, and uncensorable websites, using peer-to-peer networking to ensure that sites are hosted by multiple nodes, making it difficult to shut down [40].
- **Freenet (2000, Ian Clarke)** — Freenet is a peer-to-peer platform for censorship-resistant communication. It uses a decentralized distributed data store to keep and deliver information, supporting websites, forums, and file-sharing, emphasizing privacy and anonymity by routing data through multiple nodes [41].
- **HTTPS (Hypertext Transfer Protocol Secure) (1995, Netscape)** — HTTPS ensures secure communication over a computer network, with widespread use on the Internet. Encryption via HTTPS protects the integrity and confidentiality of data between the user's computer and the site, which is crucial for secure online transactions and data privacy [42].
- **TLS (Transport Layer Security) (1999, IETF) —**TLS is a cryptographic protocol designed to provide secure communication over a computer network. While SSL is the predecessor to TLS, both are widely used to secure segments of the internet's traffic, safeguarding data from eavesdroppers and tamperers [43].
- **SSL (Secure Sockets Layer) (1995, Netscape) —** SSL operates by establishing an encrypted link between a web server and a browser, ensuring that all data passed between them remains private and secure. This process involves using digital certificates, which authenticate the identity of the parties and enable a secure connection. The protocol uses a combination of public key and symmetric key encryption to secure connections, which helps protect against eavesdropping, tampering, and message forgery. SSL protects sensitive communications, such as online banking and shopping transactions [44].
- **IPSec (Internet Protocol Security) (1995, IETF) —** IPSec is used to secure Internet communications across an IP network by encrypting and authenticating each IP packet of a communication session. It is widely used for creating secure VPNs, particularly important for protecting data flowing between corporate networks [45].
- **SSH (Secure Shell) (1995, Tatu Ylönen) —** SSH is a protocol for operating network services securely over an unsecured network. Best known for its more secure approach to remote login than older protocols like Telnet, SSH encrypts all traffic (including passwords) to eliminate eavesdropping, connection hijacking, and other network-level attacks [46].
- **PGP (Pretty Good Privacy) (1991, Phil Zimmermann) —** Pretty Good Privacy, or PGP, is a data encryption and decryption program that provides cryptographic privacy and authentication for data communication. Developed by Phil Zimmermann in 1991, PGP is most commonly used for securing emails but is also employed to encrypt and decrypt texts, files, and even disk partitions. The system relies on symmetric-key cryptography for speed and public-key cryptography for secure key exchange, typically using the RSA algorithm. One of the key features of PGP is its use of a web of trust, where users can sign each other's keys to endorse the authenticity of the key owner's identity instead of relying on a centralized authority [47].
- **S/MIME (Secure/Multipurpose Internet Mail Extensions) (1995, RSA Data Security, Inc.) —** Secure/Multipurpose Internet Mail Extensions, or S/MIME, is a standard for public key encryption and signing of MIME data, widely used in email systems. S/MIME allows for encryption (to maintain confidentiality) and digital signatures (for authenticity and integrity). It employs a hierarchical Public Key Infrastructure (PKI) system where certificate authorities (CAs) authenticate the identity of users and devices, making it suitable for use in enterprise settings where certificates can be managed and distributed through central IT departments. S/MIME is supported by most modern email clients, making it accessible for users needing secure and verifiable email communication [48].
- **OpenPGP (1997, IETF) —** OpenPGP is an open standard for data encryption and signing, encompassing a suite of encryption, decryption, and signing algorithms. It was originally derived from the PGP software, designed to be an open standard for PGP encryption by the IETF through RFC 4880. OpenPGP is used to encrypt not only emails but also files and directories when privacy and security are required. It is unique because it allows

non-proprietary implementations, which means various software providers can implement systems that are compatible with each other. OpenPGP uses a combination of strong public-key and symmetric cryptography to ensure that communications and data, regardless of medium, remain confidential and secure. Like PGP, OpenPGP also supports the concept of a web of trust instead of relying solely on a hierarchical trust model used in S/MIME [49].

- **WPA2/3 (Wi-Fi Protected Access 2/3) - WPA2 (2004, Wi-Fi Alliance) and WPA3 (2018, Wi-Fi Alliance) —** WPA2 and WPA3 are security protocols and security certification programs developed by the Wi-Fi Alliance to secure wireless computer networks. WPA3, which replaced WPA2, provides enhanced security features, such as improved encryption and protection against brute-force attacks [50].

The Tables 3, 4, 5, 6, and 7 examine encryption techniques and highlight a substantial reliance on advanced protocols such as TOR, VPN, and others across varied computational frameworks. The deployment of such protocols has been extensively analyzed using diverse ML and DL algorithms, including Deep Neural Networks, Graph Neural Networks, and transformer-based models. These models reflect a concerted effort to tackle the complexities of detecting and classifying encrypted traffic within secure communication channels.

Despite the robust theoretical foundations of TOR and VPN technologies, practical challenges persist, particularly in their integration with ML models for real-time and accurate threat detection. The data shows a recurring theme of employing ML techniques to improve the effectiveness of these encryption services. However, the performance inconsistency across different datasets suggests a crucial challenge in generalizing these models in real-world scenarios, where encrypted traffic patterns are continuously evolving [51].

Furthermore, as digital communication becomes increasingly encrypted, maintaining privacy while ensuring security poses a dual challenge. Protocols like TOR, designed for anonymity, often complicate legitimate monitoring and threat detection tasks, leading to conflicts between privacy preservation and security enforcement [52].

Future research should, therefore, prioritize the development of adaptive encryption techniques that can operate effectively across different network architectures and conditions. Emphasizing the refinement of ML models to interpret better the nuances of encrypted traffic within TOR and VPN usages will be crucial. This includes exploring hybrid models that integrate the strengths of existing approaches to create more resilient and flexible encryption systems [53].

As the digital landscape continues evolving, so does the need for robust encryption techniques/services to safeguard sensitive information against increasingly sophisticated cyber threats. The next section transitions into a detailed literature review of technical articles, categorizing the models employed into three distinct categories: ML, DL, and Hybrid models. It examines the methodologies within each category, offering a deeper understanding of their respective strengths and limitations.

5. Related work - Technical articles

This section offers a structured overview of the methodologies employed in detecting and classifying encrypted traffic. Grounded in a comprehensive literature review, we leveraged a robust search across multiple databases, as detailed in Table 1. This search identified 60 pertinent papers published within the last five years, significantly shaping the current landscape of encrypted traffic analysis. These studies are categorized into three primary methodological approaches: ML, DL, and Hybrid Techniques. The selection and classification of these articles aligned with our search criteria to ensure comprehensive domain coverage.

In our survey, we present an expansive taxonomy of encrypted traffic classification models designed to encompass a wide spectrum of current methodologies in the domain as illustrated in Fig. 4. Our taxonomy categorizes approaches into three primary groups: Machine Learning Approaches, Deep Learning Approaches, and Hybrid Approaches. The Machine Learning tier includes foundational algorithms such as J48, Random Forest (RF), and Instance-Based Learning (IBk), which have proven essential for baseline classifications. Expanding to Deep Learning Approaches, our taxonomy captures advanced techniques ranging from basic neural networks to sophisticated configurations like Convolutional Neural Networks (CNNs) with multi-head attention and Spiking Neural Networks (SNNs), highlighting their utility in intricate feature extraction scenarios. Additionally, this group includes innovative models such as Generative Adversarial Networks (GANs) and Transformers for dynamic learning capabilities. The Hybrid Approaches layer integrates diverse methodologies, blending Machine Learning with Deep Learning (e.g., LSTM + RF + SVM + KNN) and advanced ensembles like Decision Trees (DT), RF, Multi-Layer Perceptrons (MLP) with Recursive Feature Elimination for enhanced predictive performance. This robust framework not only delineates the structured evolution of encrypted traffic analysis tools but also assesses their effectiveness across varied real-world applications, providing a detailed guide for future research directions in this rapidly evolving field.

The Tables 3,4,5,6, and 7 present an organized overview of the datasets, encryption service, algorithms, and performance metrics utilized in ML, DL, and hybrid approaches for ETC. This structured layout facilitates easy comparison across different studies and highlights the variances in methodology and results within the field.

Table 3 offers a comprehensive overview of the datasets, algorithms, and performance metrics employed in machine learning (ML) approaches for encrypted traffic classification (ETC). This table delineates the variety of datasets used, detailing their characteristics such as size, type, and the challenges they present. Additionally, it catalogs the ML algorithms applied, discussing their operational complexities and the rationale behind their selection. Performance metrics such as accuracy, precision, recall, and F1-score are systematically presented to evaluate and compare the effectiveness of each ML approach in handling encrypted traffic.

Tables 4,5,6 provides an exhaustive summary of the datasets, algorithms, and performance metrics utilized in deep learning (DL) approaches for ETC. It highlights the advanced neural network architectures and learning paradigms adopted to enhance detection capabilities. This table also examines the intricacies of dataset compatibility with DL models, emphasizing the impact of data volume and feature richness on model performance. Essential performance metrics are analyzed to gauge the efficiency and accuracy of DL techniques in various encrypted traffic scenarios.

Table 7 examines the intersection of machine learning and deep learning through hybrid approaches in encrypted traffic classification. It details the integration of ML and DL techniques to create robust models that leverage the strengths of both paradigms. This table reviews the algorithms in the context of their combined utility, showcasing how hybrid models outperform single-approach models in accuracy and reliability. Performance metrics detailed in this table highlight the synergy effects and the enhanced capability of hybrid approaches to adapt to the evolving landscape of network encryption.

These tables serve as a valuable resource for researchers, offering a clear snapshot of how different strategies perform under varied conditions and guiding future studies in selecting appropriate datasets and algorithms to advance the accuracy and efficiency of encrypted traffic detection systems.

The following subsections will delve into the diverse models detailed in our taxonomy, offering a critical analysis of each approach's efficacy. By emphasizing literature that aligns with our robust search methodology, we ensure that our survey is exhaustive and tailored to the field's most impactful and innovative studies. We further synthesize these findings into a cohesive conclusion, highlighting the present landscape and future directions in analyzing encrypted traffic detection and classification.

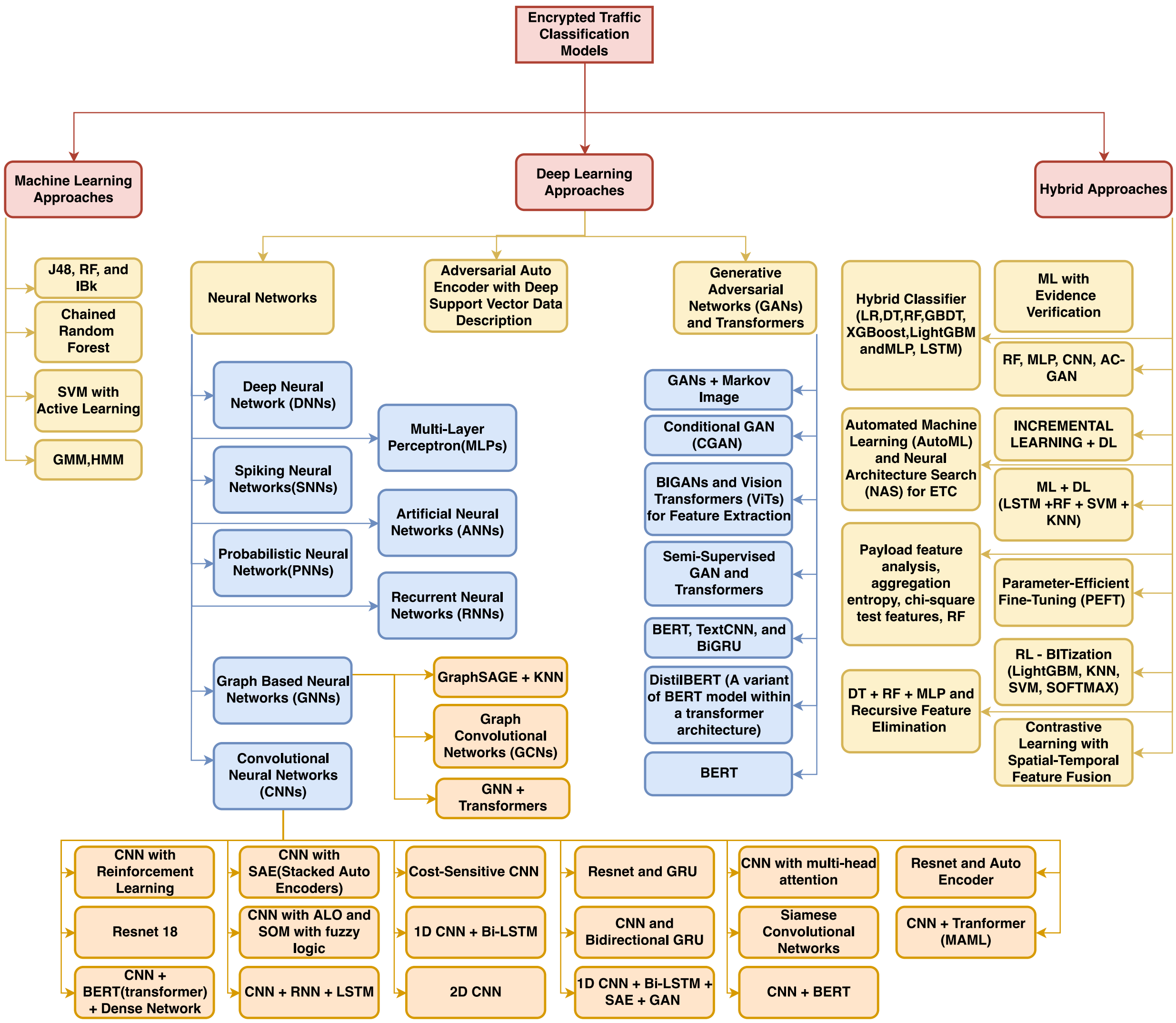

Fig. 4. Taxonomy of encrypted traffic detection models derived from previous technical literature.

5.1. ML models

In the evolving landscape of cybersecurity, ML techniques have become instrumental in decrypting the complexities of encrypted traffic [54]. This innovative research category aims to devise algorithms that can accurately classify, identify, and characterize potential cyber threats hidden within encrypted data flows [8]. The essence of these approaches lies in their ability to learn from data patterns, offering a dynamic shield against evolving network vulnerabilities [55]. We delve into the significant contributions that have pushed the boundaries of this field, each presenting a unique angle on encrypted traffic analysis.

Recent advancements in ETC leverage various innovative ML techniques to enhance the granularity and accuracy of analysis. Zaki, F. et al. developed "Granular multi-label encrypted traffic classification using classifier chain(GRAIN) [56]", which employs classifier chains to dissect network traffic into detailed classes. Their method significantly improves classification through novel statistical features derived from packet payload lengths, thereby preserving user privacy. In a similar vein, Dong, S. et al. tackled the challenge of data imbalance common in network traffic classification by introducing a cost-sensitive SVM enhanced by active learning, which dynamically adjusts weights to enhance accuracy across varied traffic types [57]. Yao, Z. et al. combined Gaussian Mixture Models with Hidden Markov Models to provide a nuanced classification of encrypted traffic, leveraging statistical and temporal patterns for improved network security analytics [58].

Furthermore, Choorod et al. utilized ML to effectively differentiate encrypted Tor and NonTor traffic. Focusing on unique patterns within encrypted payloads, their approach employs DPI and statistical analysis to offer robust tools for real-time encrypted traffic monitoring and analysis, significantly contributing to network security advancements [59].

These studies collectively underscore the potential of ML in transforming ETC, offering robust solutions to safeguard digital infrastructures.

5.2. DL models

DL approaches have significantly advanced encrypted traffic analysis in network security, offering profound insights and enhanced detection capabilities for concealed cyber threats [60]. These methodologies excel in interpreting complex data patterns, thereby fortifying digital defenses against sophisticated attacks [61]. We highlight key contributions in this area and explore the pivotal studies that have notably enriched our understanding and classification of encrypted network traffic [62].

5.2.1. Auto encoders

In encrypted traffic analysis, innovative applications of autoencoders are paving the way for enhanced precision and efficiency. Lv,

S. et al. [63] introduce Adversarial AutoEncoders with Deep Support Vector Data Description(AAE-DSVDD), a model to elevate VPN traffic analysis. This method significantly improves the analysis of VPN traffic, providing a deeper understanding of encrypted network traffic flows, thus enhancing network security measures. Additionally, Aceto, G. et al. [64] present DISTILLER, a multifaceted classifier that capitalizes on multimodal and multitask DL techniques. By harnessing diverse input data, DISTILLER effectively navigates the complexities of ETC, overcoming the constraints of traditional single-mode learning approaches. This model's versatility allows it to handle various classification tasks concurrently, showcasing a significant step forward in adapting to the dynamic nature of network security.

5.2.2. Neural networks

The exploration of neural networks in network security represents a critical evolution in identifying and classifying encrypted traffic [65]. Neural network models, with their profound learning capabilities, offer nuanced approaches to understanding complex data patterns, thereby enhancing the identification of cyber threats within encrypted data flows [66]. This section outlines significant advancements across various neural network architectures, each contributing uniquely to ETC. It includes a dedicated subsection on models integrating CNNs to enhance feature extraction and classification accuracy.

The use of neural networks in the domain of ETC has witnessed several innovative contributions that enhance network security systems' functionality and accuracy. Jorgensen, S. et al. [67] explore VPN application labeling through a Probabilistic Neural Network (PNN) that incorporates uncertainty quantification, markedly improving reliability by addressing inherent uncertainties in encrypted environments. Furthermore, Song, Z. et al. [68] introduce I^{2}RNN, an Incremental and Interpretable Recurrent Neural Network that not only focuses on interpretability but also adapts incrementally to new traffic types through a fingerprint learning process significantly enhancing the dynamics of network security measures.

Zhou, K. et al. [69] employ a combination of neural networks and entropy estimation to refine traffic classification, differentiating between encrypted and plaintext traffic while offering precise application-specific insights. Additionally, Pathmaperuma, M. H. et al. utilize Deep Neural Networks (DNNs) to achieve fine-grained detection of in-app activities, demonstrating high accuracy in identifying both known and unknown data patterns [70].

Rasteh, A. et al. explore the potential of Spiking Neural Networks (SNNs) for classifying encrypted internet traffic, capitalizing on their ability to recognize time-related data features, offering a promising alternative to conventional neural models [71]. Lastly, Xu, Y. et al. present 'FastTraffic', a lightweight Multilayer Perceptron (MLP) framework designed for real-time ETC on less capable devices, striking a balance between efficiency and efficacy [72].

5.2.2.1. Convolutional Neural Networks (CNNs). CNNs have become a cornerstone in network security, especially for the ETC [60]. By exploiting spatial hierarchies in data, CNNs offer an unparalleled ability to extract features and identify patterns within complex, encrypted data streams [73]. This segment showcases various innovative approaches leveraging CNNs, each addressing specific challenges in analyzing encrypted traffic [62].

The utilization of CNNs in the analysis of encrypted traffic has evolved with several innovative contributions enhancing the accuracy and efficiency of classification. He, Y. et al. [74] pioneered an image-based approach, transforming network session payload sizes into gray images for classification via CNNs, simplifying feature extraction, and enhancing traffic classification accuracy. Further expanding CNN applications, Moreira, R. et al. combined CNNs with Reinforcement Learning to optimize TOR traffic classification. SqueezeNet demonstrates rapid prediction capabilities vital for real-time applications [75].

Cheng, J. et al. [76] introduced MATEC, integrating multi-head attention with CNNs to excel in real-time encrypted traffic analysis, balancing computational efficiency with accuracy. Wang, M. et al. [77] and Lotfollahi, M. et al. [62] further advanced CNN utility by pairing them with Stacked Auto Encoders (SAEs), with notable success in differentiating VPN and non-VPN traffic and reducing information loss.

Soleymanpour [78] et al. addressed unbalanced data challenges in traffic classification through CSCNN, a cost-sensitive CNN model that enhances accuracy for minority classes. Concurrently, Xu, L. et al. [79] demonstrated the adaptability of Siamese convolutional networks in ETC, which can handle dataset limitations effectively.

In a novel approach, Lin, C. Y. et al. combined CNNs with Bidirectional GRUs, enhancing spatial and temporal feature extraction to analyze encrypted data streams [80] comprehensively. Izadi, S. et al. [81] proposed combining CNNs with evolutionary algorithms, such as Ant-Lion Optimizer and Self-Organizing Maps with fuzzy logic, to improve classification accuracy in cybersecurity.

Other notable contributions include Habibi Lashkari, A. et al. [82] and Lan, J. et al. [83], who enhanced darknet traffic detection using deep image learning and advanced neural mechanisms like self-attention and Bi-LSTM, respectively. Moreover, innovative frameworks like BFSN [84], AEFETA [85], and tCLD-Net [86] have utilized combinations of CNNs with LSTMs and attention mechanisms to tackle the complexities of ETC, showcasing significant enhancements in handling encrypted communications.

Further enriching the field, new architectures such as SPPNet [87], CENTIME [88], and EETC [89] leverage DL to surpass traditional classification tools inefficiencies, with focus on real-time processing and incremental learning. Also, innovative methods like the CBD model [90] and BCFNet [91] have explored the combination of CNNs with emerging technologies like BERT and multi-scale feature fusion to capture detailed traffic characteristics, thus setting new benchmarks in ETC.

5.2.2.2. Graph-based neural networks. Using Graph-Based Models in network security offers a nuanced and powerful approach to analyzing encrypted traffic. These models provide an advanced framework for detecting and classifying cyber threats by harnessing the intricate structures and relationships within network data [92,93]. This section delves into recent innovations that leverage graph-based techniques to enhance the precision and robustness of ETC [93].

Several groundbreaking approaches have been introduced in the innovative realm of graph-based models for ETC, each contributing unique insights into network traffic analysis. Diao, Z. et al. [94] have developed EC-GCN, a sophisticated framework employing multi-scale graph convolution networks to capture spatial–temporal traffic features. This dynamic, graph-based analysis adapts well to noise and changes in network conditions, marking a significant leap forward in the field.

Expanding upon this foundation, Hong, Y. et al. [95] presented 'MalDiscovery,' a graph-based method for detecting malicious traffic within encrypted communications. By integrating the GraphSAGE model with multi-view features, 'MalDiscovery' improves the detection accuracy and enhances the efficiency of analyzing malicious activities, leveraging a comprehensive graph-based approach to explore traffic correlations more deeply.

Wang, L. et al. [96] introduced TGPrint, which uses Graph Convolutional Networks (GCNs) combined with attention mechanisms to transform network traffic into attack graphs. This innovative method significantly improves the accuracy of classifying different types of attacks in encrypted traffic, including those previously unseen, showcasing the robust potential of graph-based analytical techniques.

Further advancing the field, Han et al. [97] unveiled the Dual Embedding with Graph Neural Network (DE-GNN) model, which conducts a fine-grained ETC by separating packet headers and payloads during the initial encoding phase. This method uses PacketCNN to extract packet-level features. It constructs a Traffic Interaction Graph (TIG) for analyzing flow-level features through Graph Neural Networks (GNNs), thus enhancing classification accuracy by effectively integrating these disparate data aspects.

Zhang et al. [98] proposed the Contrastive Learning Enhanced Temporal Fusion Encoder (CLE-TFE), combining supervised contrastive learning with cross-level multi-task learning for ETC. This model utilizes contrastive learning to improve semantic invariance across packet- and flow-level representations, thereby significantly boosting classification performance.

Lastly, Yang et al. [99] introduced MTSecurity, a model that uses Graph Neural Networks (GNNs) and Transformer technologies to classify encrypted malicious traffic. By combining byte features with graph-based traffic interaction features through the innovative Malicious Traffic Interaction Graph (MTIG), this model sets a new benchmark in network security analytics with its robust detection capabilities, significantly improving the classification accuracy outcomes across diverse datasets. Together, these advancements underscore the significant impact of graph-based models on the evolving landscape of ETC, highlighting their potential to transform cybersecurity practices.

5.2.3. Generative Adversarial Networks (GANs) and transformers

The category of Generative Adversarial Networks (GANs) and Transformers presents an avant-garde approach to ETC, leveraging the generative capabilities of GANs and the sophisticated processing power of transformer models [100]. These methodologies offer innovative solutions to longstanding challenges in network security, such as class imbalance, data augmentation, and feature extraction. This section showcases the notable advancements in employing GANs and transformers to analyze encrypted network traffic [100,101].

Several pioneering approaches have emerged in the domain of Generative Adversarial Networks (GANs) for the analysis of encrypted traffic. Tang, Z. et al. [102] have introduced Markov-GAN, which transforms encrypted traffic into Markov images to enhance classification, creatively merging GANs with Markov image transformation to improve dataset quality and address limitations of traditional methods. Meanwhile, Wang, P. et al. [103] tackled the issue of class imbalance through PacketCGAN, using Conditional GANs to generate samples for underrepresented classes, thereby balancing datasets and enhancing DL classifier performance in the analysis of encrypted traffic.

Continuing this innovative trend, Sanjalawe, Y. et al. [104] combined GANs with Vision Transformers (ViTs) for the detection of obfuscated internet traffic, a method that surpasses the capabilities of conventional CNNs and RNNs by leveraging GANs for data augmentation and ViTs for effective feature extraction. In a similar vein, Wang, P. et al. [105] explored ETC using the semi-supervised GAN model ByteSGAN, focusing on SDN Edge Gateways to demonstrate the adaptability of semi-supervised learning in the evolving landscape of network security. Additionally, Zhao, R. et al. [106] introduced MT-FlowFormer. This semi-supervised transformer-based framework adeptly manages the complexities of encrypted network traffic by leveraging the benefits of transformers and semi-supervised learning to address the scarcity of labeled data. Together, these studies illustrate significant advancements in applying GANs to encrypted traffic detection and classification, heralding a new era of possibilities in cybersecurity methodologies.

Further extending the use of transformers, Lin, X. et al. [107] introduced ET-BERT, employing transformers for pre-training on large-scale unlabeled encrypted traffic, which enhances traffic classification by capturing contextualized datagram representations. In a similar vein, Huang et al. [108] unveiled the BSTFNet model, which uniquely combines global semantic features extracted by transformers alongside local spatiotemporal features analyzed using BiGRU and TextCNN, effectively classifying encrypted traffic with notable accuracy on the USTC-TFC2016 dataset.

Adding to these advancements, Park et al. [109] presented a method that uses the multi-task learning model, DistilBERT, for encrypted network traffic classification. This innovative approach facilitates simultaneous training across multiple classification tasks—encapsulation, category, and application—within a single model framework, significantly enhancing efficiency and accuracy in network security management. Together, these studies showcase the significant strides in employing advanced ML techniques, particularly semi-supervised and transformer models, to tackle the complexities of ETC in the modern cybersecurity landscape.

5.3. Hybrid models

The Hybrid Model Approaches in ETC demonstrate the synergy between ML and DL and their integration with traditional methodologies. They offer robust, versatile, and innovative solutions to network security challenges [110]. This category underscores the strength of combining different analytical techniques to enhance classification accuracy, adaptability, and computational efficiency [111].

Marim, M.C. et al. [112] highlight the enduring utility of classical ML models like Decision Trees and Multilayer Perceptrons in classifying darknet traffic using the CIC-Darknet2020 dataset, emphasizing their role in offering interpretability and efficiency. Xu, B. et al. [113] introduce ME-Box, a system that integrates ML with evidence verification to balance security and privacy in encrypted traffic detection, signaling a shift towards adaptive security strategies.

Further, Hu, Y. et al. [114] detail a hierarchical hybrid classifier merging ML and DL for analyzing user behavior in darknet environments, providing deep insights into anonymous online activities. Rust-Nguyen, N. et al. [115] evaluate the resilience of ML and DL models against adversarial attacks, underscoring the necessity of robust models in cybersecurity. Malekghaini, N. et al. [116] explore AutoML and Neural Architecture Search in "AutoML4ETC", demonstrating their potential in optimizing neural networks for encrypted traffic analysis.

Elmaghraby, R. T. et al. [117] examine advanced ML and DL techniques, focusing on distinctive feature extraction such as packet length and timestamps to improve network management. Luo, P. et al. [118] propose an ML-based system incorporating 'BITization' and an optimal sliding window technique, marking significant progress in encrypted traffic pattern recognition.

Yan et al. [119] introduce a high-speed classification method using payload features and a Random Forest model, efficiently distinguishing between encrypted and unencrypted traffic. Zhao et al. [120] present METAROCKETC, a framework combining time series analysis and meta-learning for rapid adaptation to ETC tasks. This adaptive framework exemplifies integrating innovative techniques to tackle the dynamic challenges of encrypted traffic analysis.

Li et al. [121] present the MISS framework, an incremental learning system for ETC that adapts efficiently to new network applications without complete retraining. This method conserves resources and enhances model scalability by exploiting multi-view sequences to obtain a comprehensive feature set.

Wang et al. [122] propose a classification method with contrastive learning, incorporating spatial and temporal feature fusion for a consistent representation of traffic samples. Finally, Wang and Gu [123] introduce a multi-task scenario approach with a Parameter-Efficient Fine-Tuning method, enhancing computational efficiency and interpretability in ETC, potentially transforming network security practices. Together, these studies demonstrate the profound impact of hybrid models in advancing the analysis of encrypted traffic, setting new standards in network efficiency and security.

In summary, the hybrid model approaches in ETC exemplify how the convergence of ML, DL, and traditional methods can create powerful tools for addressing the complexities of network security. These integrative strategies improve the accuracy and efficiency of classification systems and pave the way for more adaptive and forward-thinking solutions in the ever-evolving field of cybersecurity.

5.4. Synthesis

Classifying encrypted traffic is a crucial yet complex challenge in the dynamic field of network security. As we delve into the multifaceted issues of computational efficiency and resource demands, this section explores the inherent limitations and potential drawbacks of current models/techniques utilized in this domain.

We focus on high computational complexity and the resource-intensive nature of preprocessing and model architecture. These factors significantly determine the practicality of deploying these models in real-time or resource-constrained environments. The limitations include but are not limited to the dependency on diverse and quality data, the scalability and complexity of models, the necessity for continuous adaptation and learning, and the feasibility of real-time processing.

Every element is essential in understanding the broader implications of ETC systems and their viability in contemporary networked environments. The following bullet points outline these limitations, each supported by specific references that, to some extent, illuminate the discussed shortcomings.

- High Computational Complexity [62,63,72,75,86,96,98,99,103,108,124]
- Resource-Intensive Preprocessing and Model Architecture [69,90,109]
- Impact of Data Quality and Diversity on Model Effectiveness [64,69,76,84,95,104,112,116,119,123]
- Challenges in Data Collection and Representation [56,74,78,107]
- Complex Model Integration and Scalability Issues [67,70,79,80,96,117,125,126]
- Scalability Concerns Due to Advanced Features [106,127]
- Need for Ongoing Model Updates [57,67,71,77,102,121,128]
- Adaptation Challenges in Evolving Network Conditions [115,120]
- Real-Time Application Feasibility [63,81,82,87–89,95,105,114]
- Practical Deployment Issues in Dynamic Networks [68,94,102]

This study of the drawbacks and limitations in the models of ETC highlights critical challenges that impact the practical deployment of sophisticated models. High computational complexity and resource-intensive processes hinder real-time applications and limit the scalability necessary for broader network implementations. By analyzing these limitations, this survey paper sets the stage for further discussion on the need for model innovation and optimization. The insights gathered here underscore the urgency for developing more efficient techniques that do not compromise performance while ensuring adaptability and minimal resource utilization. As we progress, these considerations will guide the development of more robust and practical solutions in the ever-evolving network security landscape.

6. Encrypted traffic datasets

The rapid expansion of encrypted traffic across the internet necessitates the creation of advanced techniques for its overall analysis and classification. Encrypted traffic data collection is critical in this process, providing the raw material necessary for training ML, DL, and hybrid models. The efficacy of these models heavily depends on the diversity and quality of the datasets utilized. In this section, we discuss the various encrypted traffic datasets available chronologically to underscore the characteristics and features of a high-quality dataset. We will also score the used datasets presented in recent literature and categorize them based on the encryption technique used, labels, attack diversity, etc.

- **CTU-13 (2011, CTU University, Czech Republic)** — Focuses on botnet behavior analysis in network traffic. Analyzes botnet traffic across 13 scenarios with various protocols to differentiate between malicious and benign traffic. The dataset exhibits a notable limitation in its lack of direct focus on modern encrypted traffic technologies, which may impact the generalizability of the findings [129].
- **MOORE_SET and NOC_SET(2013, Moore, A. W., & Shi Dong)** — Analyzes network behavior and security through data from internet communications at a research site and Southeast University using protocols like HTTP, FTP, and SMTP. Although valuable for network protocol analysis, the dataset falls short in representing a diverse range of global internet traffic conditions, potentially limiting its applicability in varied geographical contexts [130].
- **ISCX Tor/Non-Tor 2016 (2016, Canadian Institute for Cybersecurity)** — Includes Tor and non-Tor traffic for ETC. Mixes Tor and non-Tor traffic using protocols like HTTP, HTTPS, SMTP/S, and FTP over SSH. While the dataset effectively detects encrypted Tor traffic, its specific focus on Tor limits its broader applicability to other types of encrypted communications [131].
- **ISCX VPN/Non-VPN 2016 (2016, Canadian Institute for Cybersecurity)** — Includes traffic from VoIP, P2P, and streaming over VPN and non-VPN channels. Analyzed using ISCXFlowMeter. The dataset primarily focuses on VPN traffic, which may not adequately represent the full spectrum of network threats, thereby narrowing its utility in broader security contexts [55].
- **USTC-TFC2016 (2016, University of Science and Technology of China)** — Features traffic from applications like BitTorrent and Skype and malware communications. Aimed at distinguishing between user data and security threats. This dataset does not extensively cover all types of encrypted traffic, which may limit its effectiveness in comprehensive encrypted traffic analysis [132].
- **BetterNet HTTPS (2016, Wazen Shbair)** — Captures full HTTPS raw PCAP files from popular websites, aiding in HTTPS traffic analysis. The dataset focuses exclusively on web traffic, omitting other crucial network interactions, which might restrict its relevance in more comprehensive network studies [133].
- **Anon-17 (2017, Network Information Management and Security Lab)** — Focuses on the operational dynamics within anonymity networks such as Tor, I2P, and JonDonym. It provides an extensive collection of benign, encrypted, and non-encrypted traffic to facilitate research on the effectiveness of these anonymity services. The dataset includes detailed protocol usage across various scenarios to examine how these networks manage privacy protection. A notable limitation of Anon-17 is its exclusion of malicious traffic and modern attack vectors, which may affect the applicability of the findings to real-world threat scenarios [134,135].
- **CIC IDS2017 (2017, Canadian Institute for Cybersecurity)** — Captures benign and malicious traffic using protocols like HTTP and FTP, including DDoS and Heartbleed attacks. While the dataset is tailored towards intrusion detection, this focus might overlook the nuances of deeper encrypted traffic analysis, potentially limiting the dataset's broader applicability [136,137].
- **Google Home(2018/20, C Wang, S Kennedy)** — Focuses on device-specific interactions through 100 voice commands, collecting 1500 traffic traces per command over nine weeks. A significant dataset for studying responses within a controlled environment. The dataset is specific to Google Home, which limits its generalizability to other encrypted voice-activated devices and may reduce its applicability in broader IoT security analyses [138].
- **Mirage (2019, ARCLAB, University of Napoli 'Federico II')** — Focuses on mobile app traffic analysis by capturing encrypted traffic from Android devices. This dataset includes traffic from various applications across multiple protocols to differentiate between app-specific traffic patterns. The dataset is noted for lacking traffic diversity related to direct attacks, which may impact its applicability in network security research. However, its comprehensive data on mobile application behavior makes it invaluable for studying encrypted mobile traffic characteristics [139].

Table 3
Overview of datasets, algorithms, and performance metrics in ML approaches for ETC.

Ref.	Dataset used	Algorithm/ Model used	Type of encryption service	Precision	Recall	Accuracy	F1-Score	Others	Additional comments
Zaki et al. [56]	ISCX-VPN2016	Chained RF	VPN	100%, 100%	100%, 93%	–	100%, 94%	–	Application Name level, Application Service level
Dong [57]	MOORE_SET and NOC_SET	SVM with Active Learning	Others	–	–	–	–	G-mean: 0.718, 0.768; MAUC: 0.709, 0.774	–
Yao et al. [58]	CIC-IDS2017 and private traffic dataset collected from a self-built LAN	GMM, HMM	Others	100%	99.43%	99.98%	99.72%	–	–
Choorod et al. [59]	UNB-CIC and ISCXTor2016	J48, RF, IBk	TOR	0.93%	0.93%	95.65%	0.93%	–	Avg, J48

- **Orange (2020, Orange Labs, France)** — Focuses on encrypted web traffic classification using deep learning to enhance service detection across multiple web protocols such as HTTP/2 and QUIC. The dataset captures a comprehensive mix of encrypted traffic from a major European ISP structured to provide insights into service-level traffic behaviors without direct attack scenarios. While it offers an extensive label set for service classification, the dataset notably excludes explicit malicious traffic, which could affect its broader utility in security-focused studies. However, its detailed protocol coverage and high accuracy in service classification make it a valuable resource for developing more nuanced traffic classification systems [140].
- **CIC-Darknet-2020 (2020, Canadian Institute for Cybersecurity)** — Amalgamates traffic from Tor and VPN, analyzing activities like browsing and chat. The dataset, focused on darknet activities involving TOR and VPN, may not generalize well to other network security issues, limiting its utility in broader cybersecurity contexts [82].
- **Oregon & Ohio HTTP2 (2020, L Xu, D Dou)** — Features HTTP2 traffic from applications like Spotify and Facebook, aiming to improve ETC. The dataset is specific to HTTP2, which may hinder its generalization to analyses involving other protocols, potentially reducing its broader applicability [141].
- **TOR/I2P/Zeronet/Freenet (2020, Huy Nguyen)** — Analyzes encrypted traffic from darknet platforms, crucial for security applications. This dataset might not apply to standard network environments, which could limit its utility in typical network security assessments [142].
- **UCDavis – QUIC (2020, University of California, Davis)** — Provides real-world encrypted traffic patterns under the QUIC protocol. The dataset focuses on the QUIC protocol, potentially overlooking the nuances and implications of other network protocols, thereby limiting its comprehensive applicability [143].
- **Browser 2020 (2020, T Van Ede, R Bortolameotti)** — Offers a comprehensive view of browser-generated traffic, enhancing app detection capabilities. The dataset's focus on mobile environments may not extend its applicability to other settings, potentially limiting its relevance in diverse networking contexts [144].
- **CIC DoH Brw 2020 (2020, Canadian Institute for Cybersecurity)** — Analyzes DNS over HTTPS traffic from various browsers and servers, focusing on security against eavesdropping and man-in-the-middle attacks. The dataset is solely focused on DNS over HTTPS (DoH), which potentially overlooks a broader range of DNS security issues, thereby limiting its applicability in comprehensive DNS threat analyses [145].
- **HIKARI 2021 (2021, Ferriyan et al.)** — Focuses on encrypted synthetic attacks and normal traffic for CMSs like WordPress. The dataset utilizes Zeek for feature extraction and concentrates on specific CMS platforms, which might limit its broader application and reduce its utility in diverse cybersecurity environments [146].
- **SJTU – AN21 (2021, Shanghai Jiao Tong University)** — Analyzes anonymity network traffic from networks like Tor and I2P, aiming to enhance traffic classification. The dataset is specific to anonymity networks, which may not generalize well to standard network environments, thereby restricting its applicability in broader network security analyses [147,148].
- **CSTNET TLS 1.3 (2021, Xinjie Lin, Gang Xiong)** — Targets advanced TLS 1.3 encryption, aiding in refining traffic classification techniques. The dataset is specific to TLS 1.3, which may not cover older encryption protocols, potentially limiting its applicability in environments where legacy protocols are prevalent [107].
- **UTMobileNetTraffic2021(2021, University of Texas at Austin)** — Focuses on modern mobile traffic analysis using machine learning models to classify application-specific behaviors. The dataset contains over 29 h of encrypted traffic generated from 16 popular mobile applications, providing rich insights into application and activity-level behaviors without including direct attack scenarios. UTMobileNetTraffic2021 does not encompass any traffic related to explicit malicious activities or attack vectors, which may limit its direct applicability in security anomaly detection studies. However, with its accurate activity-level labels and diverse application coverage, this dataset is an excellent resource for developing and refining traffic classification algorithms tailored to encrypted mobile traffic [149].
- **Dataset 7cz (2022, MH Pathmaperuma, Y Rahulamathavan)** — Focuses on mobile app traffic, using advanced tools to capture detailed encrypted traffic flows. The dataset concentrates on mobile apps, potentially limiting its utility for broader network analysis and reducing its relevance in non-mobile networking contexts [70].

Table 4
Overview of datasets, algorithms, and performance metrics in DL approaches for ETC.

Ref.	Dataset used	Algorithm/ Model used	Type of encryption service	Precision	Recall	Accuracy	F1-Score	Others	Additional comments
Lv et al. [63]	ISCX-VPN2016	AAE with DSVDD	VPN	95.95%	–	–	–	AUC: 0.8682	–
Aceto et al. [64]	ISCX-VPN2016	1D CNN, BiGRU	VPN	–	–	T1: 91.95%, T2: 82.92%, T3: 66.44%	T1: 91.95%, T2: 82.92%, T3: 66.44%	–	Tasks:T1 (Encap--sulation), T2 (Traffic Type), T3 (Appli--cation)
Jorgensen et al. [67]	Self generated dataset	Prototypical networks with relative Mahalanobis distance	Others	–	–	–	0.98	–	–
Song et al. [68]	CIC-IDS 2017 and ISCXVPN 2016	RNN	VPN, Others	98.64%, 90.68%	98.59%, 91.78%	98.97%, 93.93%	98.61%, 90.13%	–	CIC-IDS2017 and ISCXVPN 2016 respectively
Zhou et al. [69]	ISCX-Tor/Non-Tor and ISCX VPN/Non-VPN	ANN	VPN, TOR	93.80%	91.20%	–	92.70%	AUC: 99%	–
Pathmape-ruma et al. [70]	Self generated, ISCX Dataset	DNN	Others	–	–	90-95%	–	–	–
Rasteh et al. [71]	CIC-Darknet-2020	SNN	VPN, TOR	–	–	95.90%	–	–	–
Xu et al. [72]	CIC-Darknet-2020, USTC-TFC 2016	MLP	TOR, VPN, Others	94.77%, 99.47%, 96.57%	94.26%, 99.33%, 94.99%	94.5%, 99.72%, 96.93%	94.4%, 99.4%, 95.53%	–	ISCXVPN-2016, ISCX-Tor 2016, USTC-TFC2016 respectively
He and Li [74]	ISCX-VPN2016	CNN	VPN	98.65%	98.64%	–	98.64%	–	–
Moreira et al. [75]	CIC-Darknet-2020	CNN with RL	VPN, TOR	–	–	99.84%	–	–	–
Wang et al. [77]	ISCX-VPN2016	CNN with SAE	VPN	98%	98%	98%	98%	–	–
Lotfollahi et al. [62]	ISCX-VPN2016	CNN with SAE	VPN	98%	93%	–	–	–	–

- **CESNET-QUIC (2022, CESNET, Czech Republic)** — Focuses on extensive QUIC protocol traffic analysis within an ISP network infrastructure. This dataset comprises one month of QUIC traffic data collected across backbone lines serving about half a million users, providing a rich basis for real-world encrypted traffic classification research. The dataset includes detailed packet metadata sequences and user agent information, facilitating in-depth studies into QUIC deployment and behavior in diverse network scenarios. A notable limitation of the CESNET-QUIC22 dataset is its focus solely on benign traffic, which may impact its use in security-focused analyses that require malicious traffic patterns. However, its comprehensive coverage of QUIC traffic and extensive metadata make it an invaluable resource for network traffic analysis and encrypted traffic studies [150].
- **AppClassNet (2022, Huawei Technologies France SASU)** — This dataset is a comprehensive tool for application identification research, modeled after the ImageNet challenges in computer vision. It comprises over 10 million flows each labeled with one of 500 fine-grained application labels using a commercial-grade DPI system. AppClassNet provides a unique resource for developing and testing machine learning models in traffic classification, particularly in real-world conditions that challenge existing methodologies. The dataset is fully encrypted and anonymized to protect user privacy and business-sensitive information, maintaining an excellent balance between utility for research and data protection. Despite these anonymizations, it preserves enough real-world characteristics to be a reliable benchmark for algorithmic research [151].

Table 5
Cont. Overview of datasets, algorithms, and performance metrics in DL approaches for ETC.

Ref.	Dataset used	Algorithm/ Model used	Type of encryption service	Precision	Recall	Accuracy	F1-Score	Others	Additional comments
Soleymanpour et al. [78]	ISCX-VPN 2016	Cost sensitive CNN	VPN	97.40%	97.40%	97.70%	97.40%	FAR: 0.026	–
Cheng et al. [76]	CTU-13, Open HTTPS, CSTNET	CNN with multi-head attention	Others	98.84%, 95.98%	98.74%, 95.53%	99.76%, 96.84%	98.79%, 95.75%	–	Open HTTPS and CSTNET respectively
Xu et al. [79]	Self-generated Private	Siamese CNN	Others	–	–	92%	–	–	–
Lin et al. [80]	ISCX-VPN 2016	CNN and Bidirectional GRU	VPN	–	93.70%	93.10%	–	–	–
Izadi et al. [81]	ISCX-VPN 2016	CNN with ALO and SOM with fuzzy logic	VPN	97%	97%	98%	–	–	–
Habibi Lashkari et al. [82]	CIC-Darknet 2020	2D-CNN	Others	86%	86%	86%	86%	–	–
Lan et al. [83]	CIC-Darknet 2020	1D CNN + Bi-LSTM	Others	–	–	92.22%	92.10%	–	–
Tong et al. [84]	ISCX-VPN 2016	1D CNN + Bi-LSTM	VPN	–	–	91%	–	–	–
Yang and Guo [85]	ISCX-VPN 2016	1D CNN + Bi-LSTM	VPN	–	98%	–	87%	–	–
Hu et al. [86]	ISCX-VPN 2016	1D CNN + Bi-LSTM	VPN	–	–	86%	–	–	–
Meslet-Millet et al. [87]	CIC-Darknet 2020	Resnet and GRU	VPN, TOR	–	–	99%	–	–	–
Maonan et al. [88]	ISCX-VPN 2016	Resnet and Auto Encoder	VPN	99.81%	99.79%	99.79%	99.80%	–	–
Ma et al. [89]	Self-Collected, ISCX-VPN 2016	Resnet 18	VPN, Others	–	–	98.1%, 96%	–	–	ISCXVPN 2016, Self-collected respectively
Tian et al. [128]	ISCX-VPN 2016, USTC-TFC 2016	Resnet 18	Others	–	–	90.55%	–	–	23,000 data samples
Seydali et al. [127]	ISCX-VPN 2016	1DCNN + Bi-LSTM + SAE + GAN	VPN	99.59%	99.44%	99.70%	99.51%	–	–

- **Pure TLS (2022, X Li, J Xie, Q Song)** — Isolates TLS encrypted traffic for detailed analysis, critical for traffic classification systems. The dataset exclusively focuses on TLS encrypted traffic, which does not accommodate analysis of non-TLS encrypted traffic, potentially limiting its comprehensiveness in encrypted traffic studies [121].
- **5GAD-2022 (2022, Cooper Coldwell)** — Aims at detecting 5G network attacks, crucial for securing 5G infrastructures. The dataset is specific to 5G networks, which may not address other types of networks and attack scenarios, limiting its applicability in broader network security assessments [152].
- **CIC IoT 2023 (2023, Canadian Institute for Cybersecurity)** — Examines network security threats for IoT devices, capturing traffic from 105 devices under 33 attack scenarios. The dataset may not comprehensively cover non-IoT network environments, which could limit its utility in broader network security studies [153].
- **VPN/NonVPN Network Application Traffic Dataset (VNAT) (2023, MIT Lincoln Laboratory)** — Includes applications like Netflix and Skype in VPN and non-VPN settings. The dataset is primarily educational and might not adequately cover industrial-scale network scenarios, potentially limiting its applicability in real-world network security environments [154].

Table 6
Cont. Overview of datasets, algorithms, and performance metrics in DL approaches for ETC.

Ref.	Dataset used	Algorithm/ Model used	Type of encryption service	Precision	Recall	Accuracy	F1-Score	Others	Additional comments
Lin et al. [125]	ISCX-Tor 2016	CNN+ RNN+ LSTM	TOR	–	–	92%	–	–	–
Hu et al. [90]	ISCX-VPN 2016	CNN+ BERT	VPN	84.49% (Macro)	83.81% (Macro)	83.88%	83.97% (Macro)	–	–
Zhu et al. [91]	ISCX-VPN 2016	ET-BERT, 1D-CNN	VPN	–	–	98.88%	–	–	–
Tang et al. [102]	USTC-TFC 2016, ISCX-VPN 2016	GANs+ Markov Image	VPN, Others	–	–	>90%	–	–	–
Wang et al. [103]	USTC-TFC 2016, ISCXVPN 2016	CGAN	VPN, Others	99.36%	99.58%	99.51%	99.47%	–	–
Sanjalawe and Fraihat [104]	ISCX-Tor 2016	BIGANs and ViTs	TOR	99.72%	99.83%	99.59%	99.78%	–	–
Wang et al. [105]	ISCX-VPN 2016	Semi-Supervised GAN and Transformers	VPN	99.16%	99.15%	99.14%	99.06%	–	–
Zhao et al. [106]	SJTU-AN21, ISCXVPN 2016	Semi-Supervised GAN and Transformers	Others	–	–	–	–	–	Outperforms state-of-the-art significantly
Lin et al. [107]	Cross-Platform (iOS& Android), USTC-TFC 2016, CIC-Darknet 2020, CSTNET-TLS 1.3	BERT	TOR, VPN, Others	–	–	–	98.9%, 92.5%, 97.4%	–	ISCX-VPN-Service, Cross -Platform (Android), CSTNET-TLS 1.3 respectively
Huang et al. [108]	USTC-TFC 2016	BSTFNet	Others	–	–	99.39%	99.40%	–	–
Park et al. [109]	ISCX-VPN 2016	DistilBERT	VPN	–	–	99.29%, 97.38%, 96.89%	–	–	Three tasks
Diao et al. [94]	OBW30 dataset, HW19 dataset	GCNs	Others	–	–	Impro--vement: 5%–20%	–	–	Over state-of-the-art methods
Hong et al. [95]	CTU-13 MCFP	Graph-SAGE+KNN	Others	99.90%	99.90%	99.90%	99.90%	–	–
Wang et al. [96]	CICIDS -2018, HIKARI -2021	GNNs	Others	99%	–	>80%	–	–	Unseen attacks
Han et al. [97]	CIC-Darknet 2020	DE-GNN	Others	94.64%	95.73%	96.09%	94.95%	–	ISCX-VPN
Zhang et al. [98]	CIC-Darknet 2020	CLE-TFE, GNN	Others	–	–	–	97.61%, 100%	–	VPN-nonVPN, Tor-nonTor respectively
Yang et al. [99]	MCFP, USTC-TFC 2016	Transformer and GNN	Others	–	–	99.46%, 99.48%	–	–	MCFP, USTC-TFC

Table 7
Overview of datasets, algorithms, and performance metrics in hybrid approaches for ETC.

Ref.	Dataset used	Algorithm/ Model used	Type of encry- -ption service	Preci- -sion	Recall	Accuracy	F1-Score	Others	Additional comments
Marim et al. [112]	CIC-Darknet2020	DT+RF+ MLP and RFE	Others	–	–	99.905%, 99.907%, 99.837%	–	–	DT, RF, MLP respe- -ctively
Xu et al. [113]	Private Continuously growing Dataset -1600 malicious flows and 1600 normal flows	ML with Evidence Verification	Others	–	–	–	–	TPR: 96.1%	–
Hu et al. [114]	Self-generated by deploying traffic collecting probes in real darknet environments to capture and label user behavior t raffic (Tor, I2P, ZeroNet, and Freenet)	Hybrid Classifier ML and DL (LR, DT, RF, GBDT, XGBoost, LightGBM, MLP, LSTM)	Others	–	–	96.9%, 91.6%	–	–	Darknet type, user behavior
Rust-Nguyen et al. [115]	CIC-Darknet 2020	RF,MLP, CNN, AC-GAN	Others	–	–	99.8%, 99.8%, 98.7%	–	–	RF,CNN, AC-GAN respectively
Malekghaini et al. [116]	Multiple Datasets-benchmark datasets and real-world TLS and QUIC traffic (Orange mobile network)	AutoML and NAS	Others	–	–	–	–	–	Outper-forms SoTA methods
Elmaghraby et al. [117]	Self-generated by performing actions on apps such as Facebook, Instagram etc.	LSTM +RF +SVM + KNN	Others	–	–	96.8%	–	–	–
Luo et al. [118]	ISCX-VPN-Service, Cross-Platform -IOS&Android, and USTC-TFC 2016	RL-BITization (LightGBM, KNN,SVM, SOFTMAX)	Others	–	–	98.60%	98.50%	–	–
Yan et al. [119]	ISCX-VPN2016, DoHBrw-2020, UNB-CIC, Wireshark capture	RF,K-means, Gaussian mixture model	Others	–	–	98.60%	94%, 85%–93%	–	Encrypted flow detection and fine-grained classifi- -cation
Zhao et al. [120]	CIC-Darknet-2020, CTU, 5GAD, Google-Home, and BetterNet -HTTPS	META- -ROCK- -ETC with MAML	VPN, Others	–	–	Impr- -ovement: 1.33%, 2.18%	–	–	5-way 1-shot and 5-way 5-shot

(continued on next page)

Table 7 (*continued*).

Li et al. [121]	Real-world network traffic datasets	MISS	Others	–	–	Impr- -ovement: 11.37%, 1.58%	–	–	–
Wang et al. [122]	ISCX-VPN 2016	CL with STFE	VPN	98.20%	98.30%	99.90%	98.20%	–	–
Wang and Gu [123]	Browser2020, CIC-IDS2017, CIC-Darknet -2020,USTC-TFC2016, CIC IoT Dataset 2023	PEFT	Others	–	–	99.98%	99.97%	–	PETR- -eLM model

- **Self Generated Private - OBW 30 HW19 (2023, Z Diao, G Xie, X Wang)** — Analyzes HTTPS traffic to classify web and application interactions, utilizing Selenium WebDriver and Scapy for data capture. Offers insights into encrypted communications. The dataset primarily focuses on HTTPS, which may not effectively capture other encrypted protocols, potentially limiting its applicability in comprehensive encrypted traffic analysis [155].
- **Self Generated Private – TCP/UDP (2024,R. T. Elmaghraby, N. M. A. Aziem)** — Includes detailed packet captures emphasizing TCP for its reliability in data delivery, with extensive classification features like IP addresses and TLS versions. Aimed at training ML models for encrypted traffic analysis. The dataset focuses predominantly on TCP traffic, possibly overlooking the nuances of UDP traffic behavior, which may limit its utility in studies requiring comprehensive protocol analysis [117].
- **BCCC-cpacket-cloud-DDoS (2024, Shafi, MohammadMoein, Arash Habibi Lashkari, Hardhik Mohanty)** — States to address challenges identified in previous datasets through a cloud infrastructure. The dataset contains over eight benign user activities and 17 DDoS attack scenarios. The dataset is fully labeled (with 26 labels) with over 300 features extracted from the network and transport layers of the traffic flows using NTLFlowLyzer. While the dataset addresses challenges from previous studies through cloud infrastructure and provides feature extraction and extensive labeling, its specific concentration on benign user activities and DDoS attack scenarios may not fully represent the broader range of security threats encountered in diverse network environments [156].

Tables 8 and 9 present the evaluation of the “31” datasets identified, drawing on related literature and research findings. Certain feature values are omitted due to incomplete metadata and documentation.

Table 10 provides a breakdown of the ‘31’ datasets, identifying each dataset by name and assigning a score according to a binary scoring system, where ‘Y’ equals 1 and ‘N’ equals 0. This scoring method allows for a standardized assessment of each dataset’s attributes, offering a clearer understanding of their suitability and completeness within the evaluated criteria.

The dataset analysis table presents an exhaustive overview of the datasets used for detecting and characterizing encrypted traffic. The table highlights key attributes such as traffic/data distribution, fully encrypted and benign/malicious traffic, and whether datasets contain encrypted and non-encrypted traffic. It also details the labeling types (binary or multi), attack diversity (e.g., Brute Force, DoS, DDoS, Web Attacks), the number of attacks/scenarios/malware, encryption services used (e.g., VPN, TOR, I2P), capture completeness, protocols (e.g., web browsing, email, VoIP), and the availability of metadata (e.g., documentation, PCAP, CSV).

From this analysis, future researchers can glean several insights. Firstly, a diverse range of encryption services and attack types are covered across these datasets, which can aid in developing robust detection and classification models. Notably, datasets including ISCX VPN/Non-VPN (2016) and VPN/NonVPN Network Application Traffic Dataset (2020) focus on VPN traffic, while others like ISCX Tor/Non-Tor (2016) and SJTU – AN21 (2021) emphasize Tor traffic. Including various protocols and attack types, such as those seen in the CICIDS (2017) and CIC IoT (2023) datasets, highlights the evolution and increasing complexity of encrypted traffic.

A critical observation from the table is the gap in datasets that combine encrypted and non-encrypted traffic with comprehensive metadata. Datasets like BetterNet HTTPS (2016) and Self-Generated Private–Pure TLS (2022) offer extensive encryption but limited metadata, which can hinder deep analysis.

Comparatively analyzing these datasets reveals distinct strengths and limitations across key attributes, which can inform dataset selection and highlight gaps in the field. For example, datasets like CICIDS (2017) and CIC IoT (2023) stand out for their extensive coverage of diverse attack types and protocol usage, making them ideal for researchers seeking comprehensive, multi-scenario data. In contrast, specialized datasets such as ISCX Tor/Non-Tor (2016) and ISCX VPN/Non-VPN (2016) are more focused, targeting specific encryption services. While this specialization can aid in targeted studies, it may limit the applicability of these datasets for broader encryption and traffic pattern analysis. Furthermore, the gap in metadata availability in certain datasets, such as BetterNet HTTPS (2016) and Self-Generated Private–Pure TLS (2022), hinders their utility for in-depth traffic characterization, where detailed contextual information is necessary. This comparative perspective suggests that while some datasets provide robust diversity in encryption and attack coverage, others are constrained by limited scope or incomplete metadata. Addressing these disparities through future dataset development could support a more comprehensive approach to encrypted traffic detection and classification, bridging the gaps in metadata and encryption service representation.

Future research in the field can learn and gain advantage from these insights by creating more datasets encompassing encrypted and non-encrypted traffic with detailed metadata. Additionally, exploring underrepresented encryption services and attack types can further enhance the understanding and capabilities of encrypted traffic detection and classification. This table is valuable for identifying current dataset limitations and guiding future development efforts in this field.

7. Information Extractors (IE) or traffic analyzers

The rise of encrypted traffic on the internet has brought significant challenges in network security and monitoring [25]. Encrypted traffic detection and classification are crucial for identifying potential security threats without compromising user privacy. Technical analyzers, also known as feature or IEs, play a pivotal role in this domain by processing traffic data and extracting relevant features for analysis [157]. These analyzers vary in approach, capabilities, and the specific features they

Table 8
Comprehensive analysis of datasets employed in ETC research.

Dataset	Traffic/Data distribution				Labeled		Attack diversity													Capture	
	Fully encrypted		Both Enc. & Non-Enc. Traffic		Binary	Multi	Name of attack												#Attacks	Complete	Partial
	Benign	Malicious	Benign	Malicious			Brute force	DoS/ DDoS	Webbased	Infiltration	C2/Botnet	Heartbleed	Reconnaissance	Spoofing	Dnsexfil.	N/W reconfig	BOT	Blockchain			
1		✗	✓	✓	✗	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	13	✓	✗
2	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
3		✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
4		✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
5		✗	✓	✓	✗	✓	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗	✓	✗	10	✓	✗
6	✓	✗		✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
7		✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
8		✗	✓	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	16	✓	✗
9		✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
10	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
11	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
12	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
13	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
14		✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
15	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
16		✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
17		✗	✓	✓	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓	✗	✗	✗	12	✓	✗
18	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
19	✓	✓		✗	✗	✓	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	3	✓	✗
20		✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
21		✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
22	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
23	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
24	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
25	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
26		✗	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗	✗	10	✓	✗
27		✗	✓	✓	✗	✓	✓	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	✗	–	✓	✗
28		✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
29	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
30	✓	✗		✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	–	✓	✗
31		✗	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	17	✓	✗

Table 9
Cont. Comprehensive analysis of datasets employed in ETC research.

Dataset	Enc. Service						Protocols														Metadata			No. of
	VPN	TOR	I2P	Zeronet	Freenet	Others	Web	Email	Chat	Stream	FTP	VOIP	P2P	DOH	5G	UDP	IRC	SSL/TLS	SSH	DB	Documentation	PCAP	CSV	#Features
1	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗	✓	✓	✗	✗	✓	✗	✓	✗	✗	✓	✓	✗	–
2	✗	✗	✗	✗	✗	✓	✓	✓	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	–
3	✗	✓	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	28
4	✓	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	24
5	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	✓	✓	✗	✓	✓	✓	✗	–
6	✗	✗	✗	✗	✗	✗	–	–	–	–	–	–	–	–	–	–	–	–	–	–	✓	✓	✗	–
7	✗	✓	✓	✗	✗	✗	✓	✓	✓	✓	✗	✓	✓	✗	✗	✓	✓	✓	✗	✗	✓	✓	✗	82
8	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	83
9	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	–
10	✓	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓	✓	✗	✓	✓	✗	133
11	✓	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗	✓	✓	✓	✗	✓	✗	✓	✗	✗	✓	✓	✗	–
12	✓	✓	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	83
13	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	–
14	✗	✓	✓	✓	✓	✗	✓	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	83
15	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✗	✗	✗	✗	✓	✓	✗	✗	✗	✓	✗	✗	–
16	✗	✗	✗	✗	✗	✗	✓	–	–	–	–	–	–	–	–	–	–	✓	–	✗	✓	✓	✗	–
17	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗	✓	✗	✓	✗	✗	✓	✓	✓	34
18	✓	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓	✗	✗	✓	✓	✗	13
19	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	✓	84
20	✗	✓	✓	✗	✗	✓	✓	✗	✓	✓	✓	✗	✓	✗	✗	✗	✓	✗	✗	✗	✓	✗	✓	83
21	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗	✓	✓	✗	✓	✓	✗	–
22	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	–	✗	✓	✗	✗	✓	✗	✓	7
23	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓	✗	✗	✓	✓	✓	24
24	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗	✓	✗	✓	✗	✗	✓	✓	✗	–
25	✗	✗	✗	✗	✗	✗	✓	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✓	✗	–
26	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗	–
27	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗	✓	✓	✓	39
28	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	✗	✗	✗	✓	✗	✗	✓	✗	✓	✓	✓	129
29	✗	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	–
30	✗	✗	✗	✗	✗	✓	✓	✗	✗	✓	✓	✓	✗	✗	✗	✓	✗	✓	✗	✗	✓	✓	✗	–
31	✗	✗	✗	✗	✗	✗	✓	✓	✗	✓	✓	✗	✗	✗	✗	✓	✗	✓	✓	✗	✓	✓	✓	348

extract, providing a range of tools for researchers and practitioners in the field [8].

This section will discuss and chronologically highlight the characteristics of various traffic analyzers or IEs that are available and widely used in research and operational domains.

- **Argus (1986, QoSient):** Argus is one of the oldest and most versatile flow monitoring tools. Its long development history has resulted in a robust tool capable of handling diverse traffic types, including encrypted flows, by generating detailed flow records [158].
- **Zeek (formerly BRO, 1995, Vern Paxson):** Zeek, the modern iteration of BRO, continues to be a leading open-source network monitoring framework. Its comprehensive scripting language and broad community support make it an invaluable tool for research and practical traffic analysis applications [159].
- **NetFlow (1996, Cisco Systems):** NetFlow, developed by Cisco, is widely employed in the industry for network traffic monitoring and analysis; it gathers IP traffic data, facilitating the identification of traffic patterns and potential security threats in both encrypted and unencrypted traffic [160].
- **Tranalyzer (2012, Tranalyzer Team):** Tranalyzer is a modular tool designed for traffic analysis and flow generation. Its plugin-based architecture allows for extensive customization, catering to specific analysis needs, including encrypted traffic [161].
- **ISCX Flowmeter(2016, Canadian Institute for Cybersecurity):** The ISCX flow meter is a comprehensive tool designed to extract

Table 10
List of datasets studied in ETC research.

S.No.	Dataset name	Reference	Score
1	CTU-13 & MCFP (Malware)-2011	CTU University [129]	16
2	MOORE_SET and NOC _SET-2013	Moore and Zuev [130]	10
3	ISCX Tor/Non-Tor-2016	Habibi Lashkari et al. [131]	14
4	ISCX VPN/Non-VPN-2016	Draper-Gil et al. [55]	17
5	USTC-TFC-2016	University of Science and Technology of China [132]	18
6	BetterNet HTTPS-2016	Shbair [133]	5
7	Anon-2017	Shahbar and Zincir-Heywood [134] and Shahbar and Zincir-Heywood [135]	17
8	CICIDS-2017	Sharafaldin et al. [136] and Shafi et al. [137]	21
9	Google Home -2018	Wang et al. [138]	5
10	Mirage-2019	Aceto et al. [139]	17
11	Orange-2020	Akbari et al. [140]	15
12	CIC-Darknet-2020	Habibi Lashkari et al. [82]	18
13	Oregon & Ohio HTTP2 Dataset-2020	Xu and Dou [141]	6
14	TOR/I2P/Zeronet/Freenet -2020	Hu et al. [142]	17
15	UCDavis–QUIC Dataset -2020	University of California, Davis [143]	11
16	Browser-2020	Van Ede and Bortolameotti [144]	7
17	CIRA-CIC-DoHBrw -2020	MontazeriShatoori et al. [145]	13
18	UTMobileNet Traffic -2021	Heng et al. [149]	16
19	HIKARI-2021	Ferriyan et al. [146]	11
20	SJTU– AN21(Tor, I2P, Jon Donym)-2021	Zhao et al. [147]	17
21	CSTNET TLS 1.3-2021	Lin et al. [107]	11
22	Self Generated Public – Dataset7cz -2021	Pathmaperuma et al. [70]	10
23	CESNET-QUIC22	Luxemburk et al. [150]	17
24	AppClassNet-2022	Wang et al. [151]	16
25	Self Generated Private – Pure TLS-2022	Li et al. [121]	9
26	5GAD dataset-2022	Coldwell et al. [152]	11
27	CIC IoT -2023	Neto et al. [153]	20
28	VPN/NonVPN Network Application Traffic Dataset (VNAT)-2023	Jorgensen et al. [154]	13
29	Self Generated Private - OBW 30 HW19-2023	Diao et al. [155]	11
30	Self Generated Private – TCP/UDP- 2024	Elmaghraby et al. [117]	12
31	BCCC-cpacket-cloud -DDoS-2024	Shafi et al. [156]	15

a broad range of features from network traffic. Its development aimed to support the needs of researchers by providing detailed flow-level information, facilitating the study of network behavior and security incidents [55].

- **CICFlowMeter (2017, Canadian Institute for Cybersecurity):** CICFlowMeter, developed by the same institute as ISCX, offers enhanced capabilities for traffic analysis. It can generate bidirectional flows with nearly 80 statistical features, crucial for analyzing encrypted and unencrypted traffic [162].
- **NFStream (2019, Free Software Foundation, Inc.):** NFStream is a multiplatform Python framework providing fast, flexible, and expressive data structures designed to make working with online or offline network data easy and intuitive. It aims to be Python's fundamental high-level building block for practical, real-world network flow data analysis. Additionally, it has the broader goal of becoming a unifying network data analytics framework for researchers providing data reproducibility across experiments. NFStream is a recent addition to the field, offering high-performance flow-based network traffic analysis. Its design prioritizes efficiency and scalability, making it suitable for modern high-speed networks [163].
- **NTLFlowLyzer (2024, MM Shafi, AH Lashkari):** NTLFlowLyzer produces bidirectional flows from the Network and Transport Layers of network traffic, with the initial packet defining the forward (source to destination) and backward (destination to source) directions. This allows for the separate calculation of statistical

time-related features in both directions. Additional capabilities include selecting from existing features, introducing new features, and managing flow timeout duration. Moreover, TCP flows are terminated upon connection teardown (by FIN or RST packet), reaching the flow's maximum duration, or being inactive for a certain amount of time (timeout) [156].

- **ALFlowLyzer (2024, MM Shafi, AH Lashkari, H Mohanty):** ALFlowLyzer generates bidirectional flows from the Application Layer of network traffic, with the initial packet defining the forward (source to destination) and backward (destination to source) directions. This enables the separate calculation of statistical time-related features for each direction. It also offers functionalities such as selecting from existing features, adding new ones, and managing the flow timeout duration. The current version supports DNS protocol; in the next versions, other protocols will be supported [164].

The Table 11 presented in this section provides a detailed examination of several prominent analyzers, including ALFlowLyzer, NTLFlowLyzer, NFStream, CICFlowMeter, ISCX Flowmeter, Tranalyzer, NetFlow, Zeek, and Argus in a chronological manner. This chronological range allows us to observe significant advancements in the number of features these tools can extract, and the programming languages used, showcasing a trend towards more robust and flexible systems. Each entry encompasses critical details such as the Analyzer's Name and Number of Extracted Features, Source code availability, Programming language, Supported protocols, and Key features they present.

The table shows a trend in the increasing complexity of supported protocols and the shift in programming languages from predominantly C in the earlier tools to Python in the more recent analyzers. For instance, 'ALFlowLyzer' (2024) and 'NFStream' (2019) utilize Python, indicating a preference for this language in contemporary network traffic analysis owing to its extensive libraries and community support. This shift also correlates with the broader implementation of ML techniques, as Python is well-suited for such applications.

The table also reveals a progression in the number of extracted features, from 30+ in 'Argus' (1986) to 348 and 130 in 'NTLFlowLyzer' (2024) and 'ALFlowLyzer' (2024) respectively. This increase is pivotal as it reflects the growing complexity of network environments and the corresponding need for more granular data to perform sophisticated analysis. For example, 'ALFlowLyzer' supports DNS protocol analysis and integrates application-layer protocol visibility, which is vital for modern cybersecurity applications.

Researchers can leverage this data to identify gaps in current methodologies or extend existing analyzers with newer protocols and ML techniques. Enhancing C-based analyzers like 'NetFlow' (1996) with modern capabilities or developing new tools that integrate cross-protocol analysis and real-time processing to address emerging security threats are viable paths forward. Additionally, open-source tools are advantageous for academic research, allowing for customization and further enhancement. This synthesis provides a comprehensive understanding of the current landscape. It underscores the continuous progression and innovation in the field, charting a path forward for the next generation of network traffic analyzers.

The broad spectrum of Traffic Analyzers or IEs available for detecting and classifying encrypted traffic demonstrates the ongoing efforts to tackle the complexities of modern network environments [165]. However, despite the advancements, there remains a notable gap in developing analyzers that can seamlessly integrate with emerging technologies and handle increasingly sophisticated encryption methods [27]. The need for new, innovative analyzers is evident, as current tools often fall short in scalability, real-time processing, and adaptability to new encryption protocols. Future research and development should focus on creating more advanced, efficient, and adaptable analyzers to meet the evolving challenges in encrypted traffic analysis.

8. Challenges & Future work

In this survey paper, we explore the various methodologies and technologies employed in detecting and classifying encrypted traffic, identifying critical gaps and highlighting the limitations of current approaches. Despite significant advances, recurring shortcomings inhibit the practical application of these technologies in real-world scenarios. The following paragraphs delineate the remaining challenges and outline future directions for research, as discussed in the subsequent sections of this paper.

In Section 3, "Previous Survey Papers", we assess the foundational insights provided by existing literature and pinpoint substantial limitations in the scope and methodologies of surveyed studies. Many focus narrowly on specific technologies, predominantly leveraging ML and DL, often overlooking their integration into real-world scenarios. This results in gaps such as inadequate protocol-specific analyses and a general disregard for operational challenges like scalability and efficiency in deployment. Future work should bridge these gaps by integrating upcoming ML/DL models with practical deployment scenarios.

Section 4, "Encryption Techniques/Services", highlights various encryption techniques' characteristics and usage patterns. Despite the wide application of encryption techniques such as VPNs and Tor, there is a lack of in-depth comparative analysis to understand why certain protocols are preferred over others. Future research should delve deeper into these choices and their implications for security solutions, aiming to improve the synergy between encryption methods and analytical models.

In Section 5, "Related Work—Technical Articles", we review the limitations and drawbacks of ETC models. Their high computational complexity and resource-intensive nature hinder their scalability and real-time application. The need for innovation in model design is clear, with future research focusing on developing more efficient methods that maintain performance while ensuring adaptability and minimal resource utilization.

Section 6, "Encrypted Traffic Datasets", examines the datasets used in encrypted traffic detection and classification. A significant challenge is the lack of datasets combining encrypted and non-encrypted traffic with comprehensive metadata. Future efforts should aim to create more inclusive datasets that cover a broader spectrum of encryption services and attack types, thus providing a richer foundation for model testing and development.

In Section 7, "Information Extractors (IE) or Traffic Analyzers", we evaluate the capabilities and limitations of various analyzers used in network traffic analysis. The current tools often fall short in scalability, real-time processing, and adaptability to new encryption protocols. Future research should focus on creating advanced analyzers that are efficient, adaptable, and capable of handling the complexities of modern encrypted traffic.

In synthesizing the findings from the various sections of this survey, the following key challenges and directions for future research are proposed, each based on specific observations and gaps identified in encrypted traffic detection and classification as denoted in Fig. 5:

- **Development of Comprehensive Models:** A critical finding across multiple studies reviewed in this survey is the need for models that can handle encrypted traffic's dynamic and evolving nature. Current machine learning (ML) and deep learning (DL) models often struggle with high-dimensional data and real-time requirements due to the computational intensity of encryption and decryption processes. Integrating advanced ML and DL techniques, such as reinforcement learning and transfer learning, has shown promise in other cybersecurity areas but remains

Table 11
Overview of analyzers/IEs employed in encrypted traffic studies.

Name of analyzer	No. of features	Source code availability	Programming language	Supported Protocols	Key features
ALFlowLyzer 2024	130	Yes	Python	DNS	Application layer protocol scalability & analysis.
NTLFlowLyzer 2024	348	Yes	Python	TCP Based	Comprehensive feature set, Impressive performance metrics, Capability to handle and analyzecloud traffic effectively
NFStream 2019	88+	Yes	Python	TCP, UDP, IP	High-performance tool for encrypted application identification and statistical feature extraction, offering extensibility via NFPlugins and a standardized framework for reproducible machine learning in network traffic management.
CICFlowMeter 2017	80	Yes	Java	TCP, UDP	Generates and analyzes bidirectional network traffic flows, calculating statistical features in both directions and offering customizable settings for feature selection and flow timeouts.
ISCX Flowmeter 2016	28	Yes	Java	TCP, UDP	Creates bidirectional flows, allowing for the separate calculation of statistical time-related features in each direction. It provides options for feature customization and flow timeout control, with flows outputting based on connection teardown for TCP and user-defined timeouts for UDP.
Tranalyzer 2008	14+	Yes	C	ARP, CDP, DNS, FTP, HTTP, ICMP, LLDP, MODBUS, MQTT, NTP, RADIUS, SSH, STP, TCP, VRRP	Lightweight, Extensible plugins/modules.
NetFlow 1996	19+	Yes	C	IP, TCP, UDP	Key features include the aggregation of packets into flows, the export of flow records to collectors, and the use of data for network monitoring and intrusion detection.
Zeek(BRO) 1995	25+	Yes	C++	Multiple protocols including HTTP, DNS, FTP	In-depth protocol analysis with application-layer insights, supported by a domain-specific scripting language for adaptable monitoring. It is designed for high-performance networks, maintaining extensive stateful data and providing a comprehensive archive of network activity.
Argus 1986	30+	Yes	C	IP, TCP, UDP, ICMP & Others	High-performance data capture, extensive protocol support, Provides a comprehensive audit engine for all network traffic, designed to support diverse network management functions including security.

Name of Analyzer: The name of the analyzer or information extractor. **No. of Features:** Tells the no. of features the analyzer extracts. **Source Code Availability:** Whether the source code for the analyzer is publicly available (Yes/No). Supported Protocols: The analyzer can process network protocols (e.g., TCP, UDP, ICMP). **Key Features:** Notable features or characteristics of the analyzer. "+" - Means protocol wise more features added(PLUGINS BASED FEATURES).

underutilized in encrypted traffic. For instance, our analysis of recent detection frameworks indicates that models trained on static datasets quickly become outdated, lacking adaptability to emerging encryption standards and novel attack patterns. Future models should simulate actual encryption scenarios, such as continuously adapting to updated encryption protocols (e.g., TLS 1.3) and evolving attack techniques, to enhance robustness and resilience. An example of this need is the integration of federated learning in privacy-sensitive environments, where centralized training can be challenging. A federated approach could enable continuous model refinement across distributed data sources without compromising user privacy.

- **Creation of Rich Datasets:** The survey identified a prominent gap in the availability and diversity of datasets covering encrypted traffic. Current datasets are limited in scope, often focusing on narrow encryption services or attack types (e.g., VPN or Tor traffic). This restricts the ability of models to generalize across different real-world scenarios. Richer datasets encompassing a wider range of encrypted services, attack vectors, and protocols are required. For example, while VPN and Tor datasets are common, there is a lack of datasets reflecting the rapidly increasing use of newer encryption methods like DNS over HTTPS (DoH) and TLS 1.3 combined with various services. Furthermore, existing datasets often lack comprehensive metadata, making it challenging to simulate real-world network conditions. As highlighted

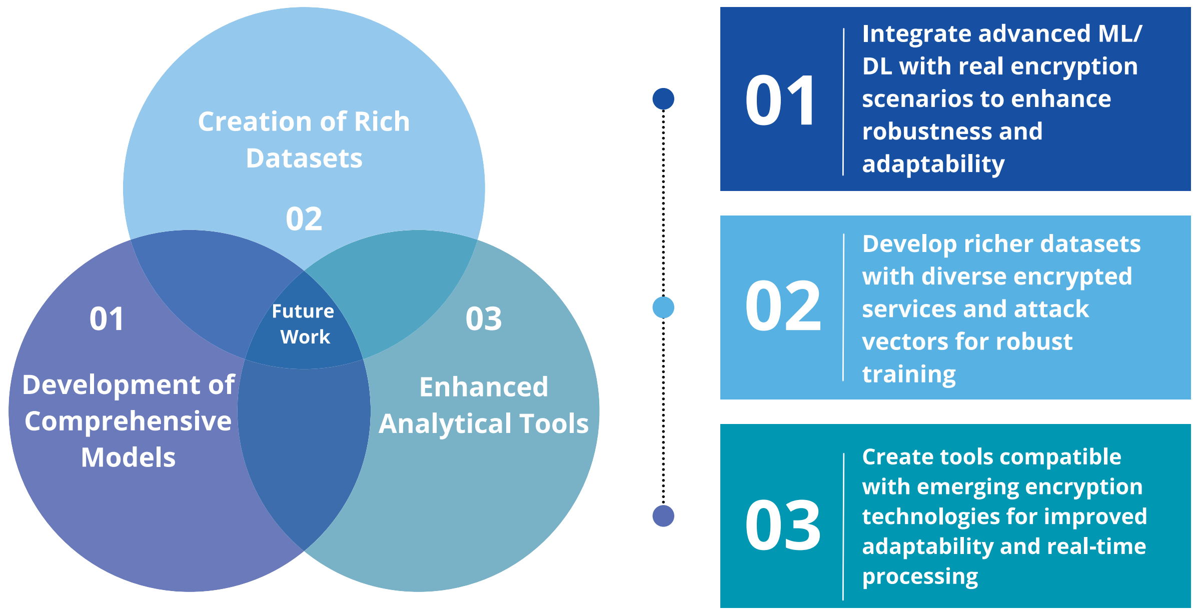

Fig. 5. Proposed future research directions.

in our dataset analysis, comprehensive datasets with labeled encrypted and non-encrypted traffic would support the development of more nuanced models capable of differentiating between legitimate and malicious traffic. Therefore, future datasets should capture more granular information, including the type of encryption and traffic context (e.g., benign vs. malicious traffic in IoT and edge devices), enabling more robust model training and validation.

- **Enhanced Analytical Tools:** Emerging encryption standards and the increasing volume of encrypted traffic have revealed limitations in existing analytical tools, particularly in real-time processing and protocol adaptability. For instance, the survey found that traditional traffic analysis tools often struggle with newer encryption technologies, which obscure payload data and thereby impede effective traffic inspection. This challenge has increased as organizations adopt stronger encryption standards like TLS 1.3, which minimizes visibility by limiting middlebox compatibility. Analytical tools need to be adaptable, efficient, and capable of functioning in real-time, with a focus on non-intrusive techniques that respect user privacy. For example, tools leveraging side-channel analysis or encrypted traffic flow characteristics could provide insights into traffic patterns without decrypting data. Enhanced tools could also use AI-driven anomaly detection to identify suspicious patterns in encrypted flows, enabling rapid responses to potential threats. To accommodate these needs, future tools should prioritize modularity, allowing them to integrate seamlessly with a variety of encryption protocols and to scale with evolving encryption methods.

This survey has highlighted various critical challenges in encrypted traffic detection and classification. Addressing these challenges through innovative research and development is necessary for advancing our capabilities in managing and safeguarding encrypted communications in a rapidly digitizing world. The emerging fields of federated learning and explainable artificial intelligence (XAI) offer promising pathways. Federated learning, particularly privacy-preserving approaches in distributed environments, demonstrates significant potential to enhance encrypted traffic classification without compromising user privacy [166–168]. Furthermore, integrating XAI can improve traffic classification models' reliability and interpretability, making them more transparent and trustworthy [169,170]. However, the adoption of these technologies in encrypted traffic classification remains limited, presenting a crucial area for future research and development. Exploring these methodologies could lead to more robust, privacy-preserving, and transparent solutions, essential for next-generation network security frameworks.

9. Conclusion

This survey has critically examined the domain of ETC, an ever-evolving field pivotal to maintaining network security in an era of ubiquitous encryption. Encryption, while essential for privacy, poses substantial hurdles to traditional network monitoring and threat detection methods, necessitating a balanced approach that respects user privacy without exposing network vulnerabilities.

Throughout this survey paper, we have traced the trajectory of advancements in the field of ETC, emphasizing the amalgamation of cutting-edge ML and DL technologies. These technologies have shown promise in parsing encrypted traffic, yet they grapple with the robustness of modern encryption and the mutable characteristics of network flow. As highlighted in the "Challenges and Future Work" section, there remains a pressing need for research strategies that refine these technologies and address operational challenges like scalability, efficiency, and adaptability to diverse encryption standards.

The analysis has revealed a significant gap in the comprehensive evaluation of encrypted traffic tools and methodologies. Many existing surveys focus too narrowly on DL and ML applications, often overlooking the practical deployment challenges critical for real-world applicability. Our findings suggest that future research should pivot towards creating more adaptive and lightweight models that maintain high performance while being feasible for everyday use. Moreover, the lack of expansive datasets that blend encrypted and non-encrypted traffic continues to be a bottleneck in developing more sophisticated analysis models.

In conclusion, while the detection and classification of encrypted traffic have advanced significantly, it faces formidable challenges that hinder its progress. This survey paper serves as a foundational resource, offering a comprehensive overview of the current state and paving the way for future research that could potentially revolutionize our ability to analyze and manage encrypted traffic effectively.

Table 12
List of acronyms/abbreviations used in the paper.

Abbreviations	Full form/Definition	Abbreviations	Full form/Definition
ETC	Encrypted Traffic Classification	ISCX	Information Security Center of Excellence
ML	Machine Learning	RADIUS	Remote Authentication Dial-In User Service
DL	Deep Learning	PCAP	Packet Capture
ETA	Encrypted Traffic Analysis	IDS	Intrusion Detection System
CNN	Convolutional Neural Network	DDoS	Distributed Denial of Service
RNN	Recurrent Neural Network	DNS	Domain Name System
IE	Information Extractor	CMS	Content Management System
IEEE	Institute of Electrical and Electronics Engineers	SJTU	Shanghai Jiao Tong University
ACM	Association for Computing Machinery	AN	Autonomous Networks
SSRN	Social Science Research Network	CSTNET	China Science and Technology Network
MDPI	Multidisciplinary Digital Publishing Institute	MIT	Massachusetts Institute of Technology
TLS	Transport Layer Security	TCP	Transmission Control Protocol
SSL	Secure Sockets Layer	UDP	User Datagram Protocol
QUIC	Quick UDP Internet Connections	BCCC	Behaviour-Centric Cybersecurity Center
DPI	Deep Packet Inspection	CSV	Comma-Separated Values
LSTM	Long Short-Term Memory	ANN	Artificial Neural Network
VPN	Virtual Private Network	RL	Reinforcement Learning
HTTPS	Hypertext Transfer Protocol Secure	BIGAN	Bidirectional Generative Adversarial Network
TOR	The Onion Router	RF	Random Forest
HTTP	Hypertext Transfer Protocol	ALO	Ant Lion Optimizer
IP	Internet Protocol	SOM	Self-Organizing Maps
SSH	Secure Shell	GMM	Gaussian Mixture Model
PGP	Pretty Good Privacy	HMM	Hidden Markov Model
MIME	Multipurpose Internet Mail Extensions	OS	Operating System
WPA	Wi-Fi Protected Access	MCFP	Multi-Class Flow Prediction
SVM	Support Vector Machine	KNN	K-Nearest Neighbors
AAE	Adversarial Autoencoder	DT	Decision Tree
DSVDD	Deep Support Vector Data Description	LR	Logistic Regression
PNN	Probabilistic Neural Network	GBDT	Gradient Boosting Decision Trees
DNN	Deep Neural Network	XGB	Extreme Gradient Boosting
SNN	Spiking Neural Network	GBM	Gradient Boosting Machine
MLP	Multilayer Perceptron	NAS	Neural Architecture Search
SAE	Stacked Autoencoder	MAUC	Mean Area Under Curve
CSCNN	Cost Sensitive Convolutional Neural Network	LAN	Local Area Network
GRU	Gated Recurrent Unit	IOS	Internet Operating System
BFSN	Binary Forward Stochastic Neurons	SOFTMAX	Softmax Function
EETC	Extended Encrypted Traffic Classification	AUC	Area Under Curve
BERT	Bidirectional Encoder Representations from Transformers	RC	Recall
GAN	Generative Adversarial Network	FAR	False Alarm Rate
CGAN	Conditional GAN	FPR	False Positive Rate
SGAN	Semi-Supervised GAN	CLETFE	Cross Layer Encrypted Traffic Feature Extraction
SDN	Software-Defined Networking	ACGAN	Auxiliary Classifier GAN
BSTFN	Bidirectional Stacked Temporal Fusion Networks	MAML	Model-Agnostic Meta-Learning
USTC	University of Science and Technology of China	FIN	Finish Flag
GCN	Graph Convolutional Network	RST	Reset Flag
PEFT	Parameter-Efficient Fine-Tuning	ALF	Application Layer Filtering
DE-GNN	Dynamic Edge Graph Neural Network	ICMP	Internet Control Message Protocol
GNN	Graph Neural Network	VRRP	Virtual Router Redundancy Protocol
TIG	Traffic Interaction Graph	ARP	Address Resolution Protocol
CLE	Contrastive Learning Enhanced	CDP	Cisco Discovery Protocol
TFE	Temporal Fusion Encoder	LLDP	Link Layer Discovery Protocol
MTIG	Malicious Traffic Interaction Graph	MQTT	Message Queuing Telemetry Transport
CIC	Canadian Institute for Cybersecurity	NTP	Network Time Protocol
ME	Machine learning and Evidence verification	CTU	Czech Technical University
STP	Spanning Tree Protocol	FTP	File Transfer Protocol
SMTP	Simple Mail Transfer Protocol		

CRediT authorship contribution statement

Adit Sharma: Writing – original draft, Visualization, Methodology, Formal analysis. **Arash Habibi Lashkari:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors report financial support was provided by Natural Sciences and Engineering Research Council of Canada (NSERC).

Acknowledgments

The authors acknowledge the grant from Canada Research Chair - Tier II (#CRC-2021-00340) and the Natural Sciences and Engineering Research Council of Canada — NSERC (#RGPIN-2020-04701) — to Arash Habibi Lashkari.

Appendix

See Table 12.

Data availability

No data was used for the research described in the article.

References

[1] A. Dainotti, C. Squarcella, E. Aben, K.C. Claffy, M. Chiesa, M. Russo, A. Pescapé, Analysis of country-wide internet outages caused by censorship, in: Proceedings of the 2011 ACM SIGCOMM Conference on Internet Measurement Conference, 2011, pp. 1–18.

[2] Google, Google Transparency Report — transparencyreport.google.com, 2024, https://transparencyreport.google.com/https/overview?hl=en.

[3] C. Wright, F. Monrose, G.M. Masson, HMM profiles for network traffic classification, in: Proceedings of the 2004 ACM Workshop on Visualization and Data Mining for Computer Security, 2004, pp. 9–15.

[4] Zscaler, Zscaler ThreatLabz 2023 state of encrypted attacks report — zscaler.com, 2023, https://www.zscaler.com/resources/2023-threatlabz-state-of-encrypted-attacks-report. (Accessed 11 November 2024).

[5] M. Handelman, USenix security '23 - rosetta: Enabling robust TLS encrypted traffic classification in diverse network environments with TCP-aware traffic augmentation — securityboulevard.com, 2023, https://securityboulevard.com/2024/01/usenix-security-23-rosetta-enabling-robust-tls-encrypted-traffic-classification-in-diverse-network-environments-with-tcp-aware-traffic-augmentation/. (Accessed 11 November 2024).

[6] ENISA, Encrypted traffic analysis: Use cases & security challenges — enisa.europa.eu, 2020, https://www.enisa.europa.eu/news/enisa-news/encrypted-traffic-analysis-use-cases-security-challenges, (Accessed 11 November 2024).

[7] M. Abbasi, A. Shahraki, A. Taherkordi, Deep learning for network traffic monitoring and analysis (NTMA): A survey, Comput. Commun. 170 (2021) 19–41.

[8] B. Anderson, S. Paul, D. McGrew, Deciphering malware's use of TLS (without decryption), J. Comput. Virol. Hack. Tech. 14 (2018) 195–211.

[9] A. Nadeem, M.Y. Javed, A performance comparison of data encryption algorithms, in: 2005 International Conference on Information and Communication Technologies, IEEE, 2005, pp. 84–89.

[10] M. Lopez-Martin, B. Carro, A. Sanchez-Esguevillas, J. Lloret, Network traffic classifier with convolutional and recurrent neural networks for Internet of Things, IEEE Access 5 (2017) 18042–18050.

[11] J. Lin, Y. Miao, L. Wei, T. Leng, K.-K.R. Choo, Efficient secure inference scheme in multiparty settings for industrial Internet of Things, IEEE Trans. Ind. Inform. (2024).

[12] J. Feng, L.T. Yang, B. Ren, D. Zou, M. Dong, S. Zhang, Tensor recurrent neural network with differential privacy, IEEE Trans. Comput. 73 (3) (2023) 683–693.

[13] P. Zhang, X. Cheng, S. Su, N. Wang, Task allocation under geo-indistinguishability via group-based noise addition, IEEE Trans. Big Data 9 (3) (2022) 860–877.

[14] L. Da Xu, Y. Lu, L. Li, Embedding blockchain technology into IoT for security: A survey, IEEE Internet Things J. 8 (13) (2021) 10452–10473.

[15] A. Reyna, C. Martín, J. Chen, E. Soler, M. Díaz, On blockchain and its integration with IoT. Challenges and opportunities, Future Gener. Comput. Syst. 88 (2018) 173–190.

[16] H.-N. Dai, Z. Zheng, Y. Zhang, Blockchain for Internet of Things: A survey, IEEE Internet Things J. 6 (5) (2019) 8076–8094.

[17] S. Rathore, Y. Pan, J.H. Park, BlockDeepNet: A blockchain-based secure deep learning for IoT network, Sustainability 11 (14) (2019) 3974.

[18] J. Feng, L.T. Yang, R. Zhang, B.S. Gavuna, Privacy-preserving tucker train decomposition over blockchain-based encrypted industrial IoT data, IEEE Trans. Ind. Inform. 17 (7) (2020) 4904–4913.

[19] K. Shahbar, A.N. Zincir-Heywood, Traffic flow analysis of tor pluggable transports, in: 2015 11th International Conference on Network and Service Management, CNSM, IEEE, 2015, pp. 178–181.

[20] K. Shahbar, A.N. Zincir-Heywood, Benchmarking two techniques for tor classification: Flow level and circuit level classification, in: 2014 IEEE Symposium on Computational Intelligence in Cyber Security, CICS, IEEE, 2014, pp. 1–8.

[21] A. Montieri, D. Ciuonzo, G. Aceto, A. Pescape, Anonymity services tor, i2p, jondonym: classifying in the dark (web), IEEE Trans. Dependable Secure Comput. 17 (3) (2018) 662–675.

[22] A. Montieri, D. Ciuonzo, G. Bovenzi, V. Persico, A. Pescapé, A dive into the dark web: Hierarchical traffic classification of anonymity tools, IEEE Trans. Netw. Sci. Eng. 7 (3) (2019) 1043–1054.

[23] T. Auld, A.W. Moore, S.F. Gull, Bayesian neural networks for internet traffic classification, IEEE Trans. Neural Netw. 18 (1) (2007) 223–239.

[24] R.K. Knake, Untangling attribution: Moving to accountability in cyberspace, in: Prepared Statement before the Subcommittee on Technology and Innovation, Committee on Science and Technology, Hearing: Planning for the Future of Cyber Attack, 2010.

[25] P. Velan, M. Čermák, P. Čeleda, M. Drašar, A survey of methods for encrypted traffic classification and analysis, Int. J. Netw. Manage. 25 (5) (2015) 355–374.

[26] M.A.I.M. Aminuddin, Z.F. Zaaba, M.K.M. Singh, D.S.M. Singh, A survey on tor encrypted traffic monitoring, Int. J. Adv. Comput. Sci. Appl. 9 (8) (2018).

[27] S. Rezaei, X. Liu, Deep learning for encrypted traffic classification: An overview, IEEE Commun. Mag. 57 (5) (2019) 76–81.

[28] A.S. Iliyasu, I. Abba, B.S. Iliyasu, A.S. Muhammad, A review of deep learning techniques for encrypted traffic classification, Unknown Journal The publication details need to be filled in as they were not provided, Unknown Year.

[29] Y. Li, H. Guo, J. Hou, Z. Zhang, T. Jiang, Z. Liu, A survey of encrypted malicious traffic detection, in: 2021 International Conference on Communications, Computing, Cybersecurity, and Informatics, CCCI, IEEE, 2021, pp. 1–7.

[30] E. Papadogiannaki, S. Ioannidis, A survey on encrypted network traffic analysis applications, techniques, and countermeasures, ACM Comput. Surv. 54 (6) (2021) 1–35.

[31] M. Shen, K. Ye, X. Liu, L. Zhu, J. Kang, S. Yu, Q. Li, K. Xu, Machine learning-powered encrypted network traffic analysis: a comprehensive survey, IEEE Commun. Surv. Tutor. (2022).

[32] D. Kahn, The Codebreakers: The Comprehensive History of Secret Communication from Ancient Times to the Internet, Simon and Schuster, 1996.

[33] R.L. Rivest, A. Shamir, L. Adleman, A method for obtaining digital signatures and public-key cryptosystems, Commun. ACM 21 (2) (1978) 120–126.

[34] W. Diffie, M.E. Hellman, New directions in cryptography, in: Democratizing Cryptography: The Work of Whitfield Diffie and Martin Hellman, 2022, pp. 365–390.

[35] X. Yi, R. Paulet, E. Bertino, X. Yi, R. Paulet, E. Bertino, Homomorphic Encryption, Springer, 2014.

[36] T. Lepistö, A. Salomaa, Automata, Languages and Programming: 15th International Colloquium, Tampere, Finland, July 11-15, 1988. Proceedings, vol. 317, Springer Science & Business Media, 1988.

[37] G. Singh-Pall, Virtual private network, 1996, https://www.microsoft.com, Developed by Microsoft Corporation.

[38] P. Syverson, M.G. Reed, D. Goldschlag, The onion router, 2002, https://www.torproject.org, Developed at the U.S. Naval Research Laboratory.

[39] Invisible internet project, 2003, https://geti2p.net, A decentralized anonymizing network.

[40] T. Kocsis, Zeronet, 2015, https://zeronet.io, Decentralized websites using Bitcoin cryptography and BitTorrent network.

[41] I. Clarke, Freenet, 2000, https://freenetproject.org, A peer-to-peer platform for censorship-resistant communication.

[42] Netscape, Hypertext transfer protocol secure, 1995, https://www.netscape.com, An extension of HTTP for secure communication over a computer network.

[43] Transport layer security, 1999, https://tools.ietf.org/html/rfc5246, Protocol for secure communications over a computer network.

[44] Netscape, Secure sockets layer, 1995, https://www.netscape.com, Protocol for establishing encrypted links between networked computers.

[45] Internet protocol security, 1995, https://tools.ietf.org/html/rfc4301, A suite of protocols for securing Internet Protocol communications.

[46] T. Ylönen, Secure shell, 1995, https://www.ssh.com/ssh/protocol/, Protocol for secure network services over an unsecured network.

[47] P. Zimmermann, Pretty good privacy, 1991, https://www.pgpi.org, Program used for encrypting and decrypting data.

[48] Secure/multipurpose internet mail extensions, 1995, https://tools.ietf.org/html/rfc5751, Standard for public key encryption and signing of MIME data.

[49] OpenPGP, 1997, https://tools.ietf.org/html/rfc4880, An open standard for encrypting and signing data.

[50] Wi-Fi protected access 2, 2004, https://www.wi-fi.org/discover-wi-fi/security, A security protocol for Wi-Fi networks.

[51] A.J. Menezes, P.C. Van Oorschot, S.A. Vanstone, Handbook of Applied Cryptography, CRC Press, 2018.

[52] M. Dworkin, Recommendation for block cipher modes of operation, NIST Spec. Publ. 800 (2001) 38B.

[53] W. Stallings, Information Privacy Engineering and Privacy by Design: Understanding Privacy Threats, Technology, and Regulations Based on Standards and Best Practices, Addison-Wesley Professional, 2019.

[54] R. Alshammari, A.N. Zincir-Heywood, Machine learning based encrypted traffic classification: Identifying ssh and skype, in: 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications, IEEE, 2009, pp. 1–8.

[55] G. Draper-Gil, A.H. Lashkari, M.S.I. Mamun, A.A. Ghorbani, Characterization of encrypted and vpn traffic using time-related, in: Proceedings of the 2nd International Conference on Information Systems Security and Privacy, ICISSP, 2016, pp. 407–414.

[56] F. Zaki, F. Afifi, S. Abd Razak, A. Gani, N.B. Anuar, GRAIN: Granular multi-label encrypted traffic classification using classifier chain, Comput. Netw. 213 (24) (2022) 109084.
[57] S. Dong, Multi class SVM algorithm with active learning for network traffic classification, Expert Syst. Appl. 176 (2021) 114885.
[58] Z. Yao, J. Ge, Y. Wu, X. Lin, R. He, Y. Ma, Encrypted traffic classification based on Gaussian mixture models and hidden Markov models, J. Netw. Comput. Appl. 166 (2020) 102711.
[59] P. Choorod, G. Weir, A. Fernando, Classifying tor traffic encrypted payload using machine learning, IEEE Access (2024).
[60] G. Aceto, D. Ciuonzo, A. Montieri, A. Pescapé, Mobile encrypted traffic classification using deep learning: Experimental evaluation, lessons learned, and challenges, IEEE Trans. Netw. Serv. Manage. 16 (2) (2019) 445–458.
[61] W. Wang, M. Zhu, J. Wang, X. Zeng, Z. Yang, End-to-end encrypted traffic classification with one-dimensional convolution neural networks, in: 2017 IEEE International Conference on Intelligence and Security Informatics, ISI, IEEE, 2017, pp. 43–48.
[62] M. Lotfollahi, M. Jafari Siavoshani, R. Shirali Hossein Zade, M. Saberian, Deep packet: A novel approach for encrypted traffic classification using deep learning, Soft Comput. 24 (3) (2020) 1999–2012.
[63] S. Lv, C. Wang, Z. Wang, S. Wang, B. Wang, Y. Zhang, AAE-DSVDD: A one-class classification model for VPN traffic identification, Comput. Netw. 236 (2023) 109990.
[64] G. Aceto, D. Ciuonzo, A. Montieri, A. Pescapé, DISTILLER: Encrypted traffic classification via multimodal multitask deep learning, J. Netw. Comput. Appl. 183 (31) (2021) 102985.
[65] Y. Zeng, H. Gu, W. Wei, Y. Guo, $Deep-Full-Range$: a deep learning based network encrypted traffic classification and intrusion detection framework, IEEE Access 7 (2019) 45182–45190.
[66] S. Rezaei, X. Liu, How to achieve high classification accuracy with just a few labels: A semi-supervised approach using sampled packets, 2018, arXiv preprint arXiv:1812.09761.
[67] S. Jorgensen, J. Holodnak, J. Dempsey, K. de Souza, A. Raghunath, V. Rivet, A. Wollaber, Extensible machine learning for encrypted network traffic application labeling via uncertainty quantification, IEEE Trans. Artif. Intell. (2023).
[68] Z. Song, Z. Zhao, F. Zhang, G. Xiong, G. Cheng, X. Zhao, B. Chen, I^2 RNN: An incremental and interpretable recurrent neural network for encrypted traffic classification, IEEE Trans. Dependable Secure Comput. (56) (2023).
[69] K. Zhou, W. Wang, C. Wu, T. Hu, Practical evaluation of encrypted traffic classification based on a combined method of entropy estimation and neural networks, ETRI J. 42 (3) (2020) 311–323.
[70] M.H. Pathmaperuma, Y. Rahulamathavan, S. Dogan, A.M. Kondoz, Deep learning for encrypted traffic classification and unknown data detection, Sensors 22 (19) (2022) 7643.
[71] A. Rasteh, F. Delpech, C. Aguilar-Melchor, R. Zimmer, S.B. Shouraki, T. Masquelier, Encrypted internet traffic classification using a supervised spiking neural network, Neurocomputing 503 (2022) 272–282.
[72] Y. Xu, J. Cao, K. Song, Q. Xiang, G. Cheng, FastTraffic: A lightweight method for encrypted traffic fast classification, Comput. Netw. 235 (2023) 109965.
[73] T. O'shea, J. Hoydis, An introduction to deep learning for the physical layer, IEEE Trans. Cogn. Commun. Netw. 3 (4) (2017) 563–575.
[74] Y. He, W. Li, Image-based encrypted traffic classification with convolution neural networks, in: 2020 IEEE Fifth International Conference on Data Science in Cyberspace, DSC, (29) IEEE, 2020, pp. 271–278.
[75] R. Moreira, L.F.R. Moreira, F. de Oliveira Silva, An intelligent network monitoring approach for online classification of darknet traffic, Comput. Electr. Eng. 110 (2023) 108852.
[76] J. Cheng, Y. Wu, E. Yuepeng, J. You, T. Li, H. Li, J. Ge, MATEC: A lightweight neural network for online encrypted traffic classification, Comput. Netw. 199 (20) (2021) 108472.
[77] M. Wang, K. Zheng, D. Luo, Y. Yang, X. Wang, An encrypted traffic classification framework based on convolutional neural networks and stacked autoencoders, in: 2020 IEEE 6th International Conference on Computer and Communications, ICCC, IEEE, 2020, pp. 634–641.
[78] S. Soleymanpour, H. Sadr, M. Nazari Soleimandarabi, CSCNN: cost-sensitive convolutional neural network for encrypted traffic classification, Neural Process. Lett. 53 (5) (2021) 3497–3523.
[79] L. Xu, D. Dou, H.J. Chao, ETCNet: encrypted traffic classification using siamese convolutional networks, in: Proceedings of the Workshop on Network Application Integration/CoDesign, 2020, pp. 51–53.
[80] C.Y. Lin, B. Chen, W. Lan, An efficient approach for encrypted traffic classification using CNN and bidirectional GRU, in: 2022 2nd International Conference on Consumer Electronics and Computer Engineering, ICCECE, (43) IEEE, 2022, pp. 368–373.
[81] S. Izadi, M. Ahmadi, R. Nikbazm, Network traffic classification using convolutional neural network and ant-lion optimization, Comput. Electr. Eng. 101 (2022) 108024.
[82] A. Habibi Lashkari, G. Kaur, A. Rahali, Didarknet: A contemporary approach to detect and characterize the darknet traffic using deep image learning, in: 2020 the 10th International Conference on Communication and Network Security, 2020, pp. 1–13.
[83] J. Lan, X. Liu, B. Li, Y. Li, T. Geng, DarknetSec: A novel self-attentive deep learning method for darknet traffic classification and application identification, Comput. Secur. 116 (2022) 102663.
[84] X. Tong, X. Tan, L. Chen, J. Yang, Q. Zheng, BFSN: a novel method of encrypted traffic classification based on bidirectional flow sequence network, in: 2020 3rd International Conference on Hot Information-Centric Networking (HotICN), IEEE, 2020, pp. 160–165.
[85] J. Yang, Y. Guo, AEFETA: Encrypted traffic classification framework based on self-learning of feature, in: 2021 6th International Conference on Intelligent Computing and Signal Processing, ICSP, IEEE, 2021, pp. 876–880.
[86] X. Hu, C. Gu, Y. Chen, F. Wei, tCLD-Net: a transfer learning internet encrypted traffic classification scheme based on convolution neural network and long short-term memory network, in: 2021 International Conference on Communications, Computing, Cybersecurity, and Informatics, CCCI, IEEE, 2021, pp. 1–5.
[87] F. Meslet-Millet, E. Chaput, S. Mouysset, SPPNet: An approach for real-time encrypted traffic classification using deep learning, in: 2021 IEEE Global Communications Conference, GLOBECOM, (44) IEEE, 2021, pp. 1–6.
[88] W. Maonan, Z. Kangfeng, X. Ning, Y. Yanqing, W. Xiujuan, CENTIME: a direct comprehensive traffic features extraction for encrypted traffic classification, in: 2021 IEEE 6th International Conference on Computer and Communication Systems, ICCCS, IEEE, 2021, pp. 490–498.
[89] X. Ma, W. Zhu, J. Wei, Y. Jin, D. Gu, R. Wang, EETC: An extended encrypted traffic classification algorithm based on variant resnet network, Comput. Secur. 128 (49) (2023) 103175.
[90] X. Hu, C. Gu, Y. Chen, F. Wei, CBD: A deep-learning-based scheme for encrypted traffic classification with a general pre-training method, Sensors 21 (24) (2021) 8231.
[91] P. Zhu, G. Wang, J. He, Y. Dong, Y. Chang, An encrypted traffic identification method based on multi-scale feature fusion, Array (2024) 100338.
[92] M. Shen, J. Zhang, L. Zhu, K. Xu, X. Du, Accurate decentralized application identification via encrypted traffic analysis using graph neural networks, IEEE Trans. Inf. Forensics Secur. 16 (2021) 2367–2380.
[93] T.-L. Huoh, Y. Luo, P. Li, T. Zhang, Flow-based encrypted network traffic classification with graph neural networks, IEEE Trans. Netw. Serv. Manag. 20 (2) (2022) 1224–1237.
[94] Z. Diao, G. Xie, X. Wang, R. Ren, X. Meng, G. Zhang, M. Qiao, EC-GCN: A encrypted traffic classification framework based on multi-scale graph convolution networks, Comput. Netw. 224 (57) (2023) 109614.
[95] Y. Hong, Q. Li, Y. Yang, M. Shen, Graph based encrypted malicious traffic detection with hybrid analysis of multi-view features, Inform. Sci. (2023) 119229.
[96] L. Wang, X. Ma, N. Li, Q. Lv, Y. Wang, W. Huang, H. Chen, TGPrint: Attack fingerprint classification on encrypted network traffic based graph convolution attention networks, Comput. Secur. 135 (2023) 103466.
[97] X. Han, G. Xu, M. Zhang, Z. Yang, Z. Yu, W. Huang, C. Meng, DE-GNN: Dual embedding with graph neural network for fine-grained encrypted traffic classification, Comput. Netw. 245 (2024) 110372.
[98] H. Zhang, X. Xiao, L. Yu, Q. Li, Z. Ling, Y. Zhang, One train for two tasks: An encrypted traffic classification framework using supervised contrastive learning, 2024, arXiv preprint arXiv:2402.07501.
[99] J. Yang, X. Jiang, Y. Lei, W. Liang, Z. Ma, S. Li, MTSecurity: Privacy-preserving malicious traffic classification using graph neural network and transformer, IEEE Trans. Netw. Serv. Manag. (2024).
[100] D.A. Hudson, L. Zitnick, Generative adversarial transformers, in: International Conference on Machine Learning, PMLR, 2021, pp. 4487–4499.
[101] J. Zhai, P. Lin, Y. Cui, L. Xu, M. Liu, GraphCWGAN-GP: A novel data augmenting approach for imbalanced encrypted traffic classification., CMES Comput. Model. Eng. Sci. 136 (2) (2023).
[102] Z. Tang, J. Wang, B. Yuan, H. Li, J. Zhang, H. Wang, Markov-GAN: Markov image enhancement method for malicious encrypted traffic classification, IET Inf. Secur. 16 (6) (2022) 442–458.
[103] P. Wang, S. Li, F. Ye, Z. Wang, M. Zhang, PacketCGAN: Exploratory study of class imbalance for encrypted traffic classification using CGAN, in: ICC 2020-2020 IEEE International Conference on Communications, ICC, IEEE, 2020, pp. 1–7.
[104] Y. Sanjalawe, S. Fraihat, Detection of obfuscated tor traffic based on bidirectional generative adversarial networks and vision transform, Comput. Secur. 135 (2023) 103512.
[105] P. Wang, Z. Wang, F. Ye, X. Chen, Bytesgan: A semi-supervised generative adversarial network for encrypted traffic classification in SDN edge gateway, Comput. Netw. 200 (39) (2021) 108535.
[106] R. Zhao, X. Deng, Z. Yan, J. Ma, Z. Xue, Y. Wang, MT-FlowFormer: A semi-supervised flow transformer for encrypted traffic classification, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 2576–2584.
[107] X. Lin, G. Xiong, G. Gou, Z. Li, J. Shi, J. Yu, Et-bert: A contextualized datagram representation with pre-training transformers for encrypted traffic classification, in: Proceedings of the ACM Web Conference 2022, 2022, pp. 633–642.

[108] H. Huang, X. Zhang, Y. Lu, Z. Li, S. Zhou, BSTFNet: An encrypted malicious traffic classification method integrating global semantic and spatiotemporal features., Comput. Mater. Contin. 78 (3) (2024).
[109] J.-T. Park, C.-Y. Shin, U.-J. Baek, M.-S. Kim, Fast and accurate multi-task learning for encrypted network traffic classification, Appl. Sci. 14 (7) (2024) 3073.
[110] X. Yun, Y. Wang, Y. Zhang, C. Zhao, Z. Zhao, Encrypted TLS traffic classification on cloud platforms, IEEE/ACM Trans. Netw. 31 (1) (2022) 164–177.
[111] S. Disabato, M. Roveri, C. Alippi, Distributed deep convolutional neural networks for the internet-of-things, IEEE Trans. Comput. 70 (8) (2021) 1239–1252.
[112] M.C. Marim, P.V.B. Ramos, A.B. Vieira, A. Galletta, M. Villari, R.M. de Oliveira, E.F. Silva, Darknet traffic detection and characterization with models based on decision trees and neural networks, Intell. Syst. Appl. 18 (2023) 200199.
[113] B. Xu, G. He, H. Zhu, ME-Box: A reliable method to detect malicious encrypted traffic, J. Inf. Secur. Appl. 59 (2021) 102823.
[114] Y. Hu, F. Zou, L. Li, P. Yi, Traffic classification of user behaviors in tor, i2p, zeronet, freenet, in: 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), IEEE, 2020, pp. 418–424.
[115] N. Rust-Nguyen, S. Sharma, M. Stamp, Darknet traffic classification and adversarial attacks using machine learning, Comput. Secur. 127 (2023) 103098.
[116] N. Malekghaini, E. Akbari, M.A. Salahuddin, N. Limam, R. Boutaba, B. Mathieu, S. Tuffin, AutoML4ETC: Automated neural architecture search for real-world encrypted traffic classification, IEEE Trans. Netw. Serv. Manag. (23) (2023).
[117] R.T. Elmaghraby, N.M.A. Aziem, M.A. Sobh, A.M. Bahaa-Eldin, Encrypted network traffic classification based on machine learning, Ain Shams Eng. J. 15 (2) (2024) 102361.
[118] P. Luo, J. Chu, G. Yang, IP packet-level encrypted traffic classification using machine learning with a light weight feature engineering method, J. Inf. Secur. Appl. 75 (2023) 103519.
[119] X. Yan, L. He, Y. Xu, J. Cao, L. Wang, G. Xie, High-speed encrypted traffic classification by using payload features, Digit. Commun. Netw. (2024).
[120] J. Zhao, Q. Li, Y. Hong, M. Shen, MetaRockETC: Adaptive encrypted traffic classification in complex network environments via time series analysis and meta-learning, IEEE Trans. Netw. Serv. Manag. (55) (2024).
[121] X. Li, J. Xie, Q. Song, Y. Sang, Y. Zhang, S. Li, T. Zang, Let model keep evolving: Incremental learning for encrypted traffic classification, Comput. Secur. 137 (2024) 103624.
[122] R. Wang, Z. Wu, Y. Li, F. Li, S. Tian, J. Liu, Encrypted traffic classification based on contrastive learning with spatial-temporal feature fusion, in: International Conference on Computer Application and Information Security (ICCAIS 2023), Vol. 13090, SPIE, 2024, pp. 531–538.
[123] G. Wang, Y. Gu, Multi-task scenario encrypted traffic classification and parameter analysis, Sensors 24 (10) (2024) 3078.
[124] R. Xie, Y. Wang, J. Cao, E. Dong, M. Xu, K. Sun, M. Zhang, Rosetta: Enabling robust tls encrypted traffic classification in diverse network environments with tcp-aware traffic augmentation, in: Proceedings of the ACM Turing Award Celebration Conference-China 2023, 2023, pp. 131–132.
[125] K. Lin, X. Xu, H. Gao, TSCRNN: A novel classification scheme of encrypted traffic based on flow spatiotemporal features for efficient management of IIoT, Comput. Netw. 190 (41) (2021) 107974.
[126] Z. Chen, G. Cheng, Z. Wei, D. Niu, Classify traffic rather than flow: Versatile multi-flow encrypted traffic classification with flow clustering, IEEE Trans. Netw. Serv. Manag. (38) (2023).
[127] M. Seydali, F. Khunjush, B. Akbari, J. Dogani, CBS: A deep learning approach for encrypted traffic classification with mixed spatio-temporal and statistical features, IEEE Access (36) (2023).
[128] S. Tian, Y. Gao, G. Yuan, R. Zhang, J. Zhao, S. Zhang, An encrypted traffic classification method based on contrastive learning, in: Proceedings of the 8th International Conference on Communication and Information Processing, 2022, pp. 101–105.
[129] CTU University, CTU-13 dataset, 2011, URL https://www.stratosphereips.org/datasets-ctu13.
[130] A.W. Moore, D. Zuev, The art of network traffic classification: 10 years after, ACM SIGCOMM Comput. Commun. Rev. 35 (3) (2005) 133–146, URL https://dl.acm.org/doi/10.1145/1070873.1070877.
[131] A. Habibi Lashkari, G. Draper-Gil, M. Mamun, A. Ghorbani, Characterization of tor traffic using time based features, in: International Conference on Information Systems Security and Privacy, 2017.
[132] University of Science and Technology of China, USTC-TFC2016, 2016, URL https://github.com/yungshenglu/USTC-TFC2016/tree/master/Benign.
[133] W. Shbair, BetterNet HTTPS, 2016, URL https://betternet.lhs.loria.fr/datasets/https/.
[134] K. Shahbar, A.N. Zincir-Heywood, Anon17: Network Traffic Dataset of Anonymity Services, Tech. Rep, Faculty of Computer Science Dalhousie University, 2017.
[135] K. Shahbar, A.N. Zincir-Heywood, How far can we push flow analysis to identify encrypted anonymity network traffic? in: NOMS 2018-2018 IEEE/IFIP Network Operations and Management Symposium, IEEE, 2018, pp. 1–6.
[136] I. Sharafaldin, A.H. Lashkari, A.A. Ghorbani, et al., Toward generating a new intrusion detection dataset and intrusion traffic characterization, ICISSp 1 (2018) 108–116.
[137] M. Shafi, A.H. Lashkari, A.H. Roudsari, NLFlowLyzer: Toward generating an intrusion detection dataset and intruders behavior profiling through network layer traffic analysis and pattern extraction, Comput. Secur. (2024) 104160.
[138] C. Wang, S. Kennedy, H. Li, K. Hudson, G. Atluri, X. Wei, W. Sun, B. Wang, Fingerprinting encrypted voice traffic on smart speakers with deep learning, in: Proceedings of the 13th ACM Conference on Security and Privacy in Wireless and Mobile Networks, 2020, pp. 254–265.
[139] G. Aceto, D. Ciuonzo, A. Montieri, V. Persico, A. Pescapè, MIRAGE: Mobile-app traffic capture and ground-truth creation, 2019, http://dx.doi.org/10.21227/maj9-vh13.
[140] I. Akbari, M.A. Salahuddin, L. Ven, N. Limam, R. Boutaba, B. Mathieu, S. Moteau, S. Tuffin, A look behind the curtain: traffic classification in an increasingly encrypted web, in: Proceedings of the ACM on Measurement and Analysis of Computing Systems, 5, 2021, pp. 1–26.
[141] L. Xu, D. Dou, Oregon & Ohio HTTP2, 2020, URL https://drive.google.com/drive/folders/1CHKcWotJg_jjE2HH6g-Z1lfr1RBBy0LO.
[142] Y. Hu, F. Zou, L. Li, P. Yi, Traffic classification of user behaviors in Tor, I2P, ZeroNet, Freenet, in: 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2020, pp. 418–424, http://dx.doi.org/10.1109/TrustCom50675.2020.00064.
[143] University of California, Davis, UCDavis – QUIC, 2020, URL https://www.kaggle.com/datasets/guillaumefraysse/ucdavisquic.
[144] T. Van Ede, R. Bortolameotti, Browser 2020, 2020, URL https://drive.google.com/file/d/1wOdrfazbrcMDrL0NfA4GLoWegtPqkPj3/view.
[145] M. MontazeriShatoori, L. Davidson, G. Kaur, A.H. Lashkari, Detection of doh tunnels using time-series classification of encrypted traffic, in: 2020 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCom/CyberSciTech), IEEE, 2020, pp. 63–70.
[146] A. Ferriyan, A.H. Thamrin, K. Takeda, J. Murai, Generating network intrusion detection dataset based on real and encrypted synthetic attack traffic, Appl. Sci. 11 (17) (2021) http://dx.doi.org/10.3390/app11177868, URL https://www.mdpi.com/2076-3417/11/17/7868.
[147] R. Zhao, Y. Huang, X. Deng, Z. Xue, J. Li, Z. Huang, Y. Wang, Flow transformer: A novel anonymity network traffic classifier with attention mechanism, in: 17th International Conference on Mobility, Sensing and Networking, MSN, 2021, pp. 223–230, http://dx.doi.org/10.1109/MSN53354.2021.00045.
[148] R. Zhao, X. Deng, Y. Wang, L. Chen, M. Liu, Z. Xue, Y. Wang, Flow sequence-based anonymity network traffic identification with residual graph convolutional networks, in: IEEE/ACM International Symposium on Quality of Service (IWQoS), 2022, pp. 1–10.
[149] Y. Heng, V. Chandrasekhar, J.G. Andrews, UTMobileNetTraffic2021: A labeled public network traffic dataset, IEEE Netw. Lett. 3 (3) (2021) 156–160.
[150] J. Luxemburk, K. Hynek, T. Čejka, A. Lukačovič, P. Šiška, CESNET-QUIC22: A large one-month QUIC network traffic dataset from backbone lines, Data Brief 46 (2023) 108888.
[151] C. Wang, A. Finamore, L. Yang, K. Fauvel, D. Rossi, AppClassNet: A commercial-grade dataset for application identification research, ACM SIGCOMM Comput. Commun. Rev. 52 (3) (2022) 19–27.
[152] C. Coldwell, D. Conger, E. Goodell, B. Jacobson, B. Petersen, D. Spencer, M. Anderson, M. Sgambati, Machine learning 5G attack detection in programmable logic, in: 2022 IEEE Globecom Workshops (GC Wkshps), 2022, pp. 1365–1370, http://dx.doi.org/10.1109/GCWkshps56602.2022.10008647.
[153] E.C.P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, A.A. Ghorbani, CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment, Sensors 23 (13) (2023) 5941.
[154] S. Jorgensen, J. Holodnak, J. Dempsey, K. de Souza, A. Raghunath, V. Rivet, N. DeMoes, A. Alejos, A. Wollaber, Extensible machine learning for encrypted network traffic application labeling via uncertainty quantification, IEEE Trans. Artif. Intell. 5 (1) (2024) 420–433, http://dx.doi.org/10.1109/TAI.2023.3244168.
[155] Z. Diao, G. Xie, X. Wang, R. Ren, X. Meng, G. Zhang, K. Xie, M. Qiao, EC-GCN: A encrypted traffic classification framework based on multi-scale graph convolution networks, Comput. Netw. 224 (2023) 109614.
[156] M. Shafi, A.H. Lashkari, V. Rodriguez, R. Nevo, Toward generating a new cloud-based Distributed Denial of Service (DDoS) dataset and cloud intrusion traffic characterization, Information 15 (4) (2024) 195.
[157] C.V. Wright, F. Monrose, G.M. Masson, On inferring application protocol behaviors in encrypted network traffic, J. Mach. Learn. Res. 7 (12) (2006).
[158] C. Bullard, Openargus - Home — openargus.org, 1984, https://openargus.org/. (Accessed 25 July 2024).
[159] V. Paxton, 1995. https://zeek.org/, (Accessed 25 July 2024).
[160] K. Delgadillo, C.I.P. Marketing, Netflow Services and Applications, Cisco Whitepaper, 1996.
[161] T. Team, Lightweight flow generator and packet analyzer - Tranalyzer — tranalyzer.com, 2012, https://tranalyzer.com/about#theanteater. (Accessed 25 July 2024).

[162] A.H. Lashkari, G.D. Gil, M.S.I. Mamun, A.A. Ghorbani, Characterization of tor traffic using time based features, in: International Conference on Information Systems Security and Privacy, Vol. 2, SciTePress, 2017, pp. 253–262.

[163] Z. Aouini, A. Pekar, NFStream: A flexible network data analysis framework, Comput. Netw. 204 (2022) 108719, http://dx.doi.org/10.1016/j.comnet.2021.108719, URL https://www.sciencedirect.com/science/article/pii/S1389128621005739.

[164] M. Shafi, A.H. Lashkari, H. Mohanty, Unveiling malicious DNS behavior profiling and generating benchmark dataset through application layer traffic analysis, Comput. Electr. Eng. 118 (2024) 109436.

[165] T.T. Nguyen, G. Armitage, A survey of techniques for internet traffic classification using machine learning, IEEE Commun. Surv. Tutor. 10 (4) (2008) 56–76.

[166] X. Yan, Y. Miao, X. Li, K.-K.R. Choo, X. Meng, R.H. Deng, Privacy-preserving asynchronous federated learning framework in distributed iot, IEEE Internet Things J. 10 (15) (2023) 13281–13291.

[167] Y. Lu, X. Huang, Y. Dai, S. Maharjan, Y. Zhang, Blockchain and federated learning for privacy-preserved data sharing in industrial IoT, IEEE Trans. Ind. Inform. 16 (6) (2019) 4177–4186.

[168] Z. Jin, K. Duan, C. Chen, M. He, S. Jiang, H. Xue, FedETC: Encrypted traffic classification based on federated learning, Heliyon 10 (16) (2024).

[169] A. Nascita, A. Montieri, G. Aceto, D. Ciuonzo, V. Persico, A. Pescapé, XAI meets mobile traffic classification: Understanding and improving multimodal deep learning architectures, IEEE Trans. Netw. Serv. Manag. 18 (4) (2021) 4225–4246.

[170] A. Nascita, A. Montieri, G. Aceto, D. Ciuonzo, V. Persico, A. Pescapé, Improving performance, reliability, and feasibility in multimodal multitask traffic classification with XAI, IEEE Trans. Netw. Serv. Manag. 20 (2) (2023) 1267–1289.

Adit Sharma is a Research Assistant at the Behaviour-Centric Cybersecurity Center(BCCC)at York University, where he is currently focused on encrypted traffic detection and classification under the supervision of Prof. Arash Habibi Lashkari. As a Graduate Student in Information Technology, Adit is actively engaged in cybersecurity research while also working as a Teaching Assistant for computer programming courses. Previously, he worked as a Senior Research and Analytics Officer at Broadband India Forum and gained experience as an intern at the European Business and Technology Centre. Adit graduated with a Bachelor of Technology in Computer Engineering from Manipal University Jaipur, where he achieved a CGPA of 9.07, and is now pursuing his Master's degree at York University.

Dr. Arash Habibi Lashkari, a Canada Research Chair (CRC) in Cybersecurity, holds a prominent position as an Associate Professor at the School of Information Technology at York University. As the founder and director of the Behaviour-Centric Cybersecurity Center (BCCC), with an extensive background spanning over 26 years in industry and academia, he has taught and conducted research & development at various international universities and organizations, contributing significantly to the field. Dr. Lashkari's expertise has earned him numerous accolades, including 15 international cybersecurity competition awards and three gold awards. He was also recognized among Canada's Top 150 Researchers in 2017. With a remarkable publication record, including 11 books and over 120 academic articles, his work covers diverse cybersecurity topics. He focuses on developing vulnerability detection technology to safeguard network systems against cyberattacks. He also has extensive industrial and development experience in network, software, information, and computer security.